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CHAPTER 12

RADIOACTIVITY

SUGGESTED TIME: 2 PERIODS

IN THIS CHAPTER

This chapter studies the stability of atomic nuclei, the nature and properties of radioactive radiations (alpha, beta, gamma), and the energy released in nuclear processes through the mass-energy relation. It ends with the two great nuclear transformations that power reactors and stars: fission and fusion.

- By the end of this chapter I can write the symbolic representation of a nuclide and identify isotopes
- By the end of this chapter I can explain nuclear stability using the N/Z ratio and the valley of stability
- By the end of this chapter I can write and balance nuclear equations using conservation of nucleon number A and charge number Z
- By the end of this chapter I can state the nature, charge, mass, penetrating power and ionising power of alpha, beta-minus, beta-plus and gamma radiations
- By the end of this chapter I can describe practical applications of radioactivity in medicine, agriculture, industry and dating
- By the end of this chapter I can define and calculate the mass defect and binding energy of a nucleus

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THIS SIGNED COURSE CONTAINS

COURSE complete, illustrated, step by step

METHODS and worked examples

EXERCISES exam style + solutions

SUMMARY SHEET the essentials to remember

CONTENTS

1. Atomic Nucleus and Nuclear Stability
2. Types of Radiation, Properties and Applications
3. Mass Defect and Nuclear Energy
4. Nuclear Fission and Nuclear Fusion
5. Integration activity
6. Methods and worked examples
7. Exercises
8. Solutions to the exercises
9. Summary sheet

CHAPTER OBJECTIVES

- By the end of this chapter I can write the symbolic representation of a nuclide and identify isotopes
- By the end of this chapter I can explain nuclear stability using the N/Z ratio and the valley of stability
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- By the end of this chapter I can describe practical applications of radioactivity in medicine, agriculture, industry and dating
- By the end of this chapter I can define and calculate the mass defect and binding energy of a nucleus
- By the end of this chapter I can apply Einstein's relation $E = mc^2$ to compute energy released in nuclear reactions (in joules and in MeV)
- By the end of this chapter I can use the binding energy per nucleon curve to predict whether a reaction releases or absorbs energy
- By the end of this chapter I can distinguish nuclear fission from nuclear fusion, write typical equations, and give their conditions and applications

PREREQUISITES

- Structure of the atom: protons, neutrons, electrons; atomic number Z and mass number A (Lower Sixth)
- Conservation of electric charge and of matter in chemical/physical equations
- Energy units: joule, electron-volt; conversions and powers of ten
- Mass-energy equivalence idea $E = mc^2$ introduced qualitatively in Lower Sixth
- Exponential behaviour and half-life notions (radioactive decay law from Lower Sixth, useful for dating applications)

1 ATOMIC NUCLEUS AND NUCLEAR STABILITY

In 2019, a cancer patient at the Yaoundé General Hospital received radiotherapy: a tiny radioactive source, no bigger than a grain of rice, destroyed tumour cells without a single surgical cut. Yet the same word "radiation" makes market traders in Douala panic when they hear about uranium mining projects in the North Region. How can the same physical phenomenon both heal and harm? And how can a few grams of matter release, through fission or fusion, the energy of tonnes of petrol? The answer lies hidden in a region of space a hundred thousand times smaller than the atom itself: the **nucleus**. In this first section we open the nucleus, identify its inhabitants, and discover why some nuclei sit quietly forever while others spontaneously explode, emitting the radiations that can kill cancer cells — or healthy ones.

1.1 Composition of the Nucleus

Observation. Every atom around you — the carbon in your exercise book, the oxygen you breathe, the iron in a Douala–Yaoundé railway line — is built on the same plan: a tiny, dense, positively charged central **nucleus**, surrounded by a cloud of electrons. The nucleus occupies almost no space (its radius is of order 10^{-15} m while the atom's is of order 10^{-10} m), yet it contains more than 99.97 % of the atom's mass. Whatever makes atoms heavy, and whatever makes some of them unstable, lives in the nucleus.

The nucleus is made of two kinds of particles, collectively called **nucleons**:

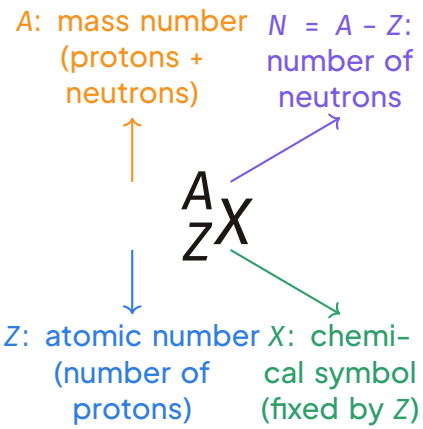
- the **proton**, of charge $+e = +1.60 \times 10^{-19}$ C and mass $m_p = 1.673 \times 10^{-27}$ kg;
- the **neutron**, electrically neutral, of mass $m_n = 1.675 \times 10^{-27}$ kg, very slightly heavier than the proton.

DEFINITION **Nucleon, nuclide, isotope**

A **nucleon** is any particle found in the nucleus: a proton or a neutron.
A **nuclide** is a nuclear species characterised by its number Z of protons (**atomic number**) and its total number A of nucleons (**mass number**). It is written symbolically as

$$^{A}_{Z}X \qquad \text{with} \qquad N = A - Z \text{ neutrons},$$

where X is the chemical symbol of the element.
Isotopes are nuclides of the same element (same Z) but with different numbers of neutrons (different A).



Anatomy of the nuclide symbol ${}^{A}_{Z}X$: Z fixes the element, A counts all nucleons, and $N = A - Z$ counts the neutrons.

EXEMPLE **Reading nuclide symbols**

- ${}^{12}_{6}\mathrm{C}$: $Z = 6$ protons, $A = 12$ nucleons, so $N = 12 - 6 = 6$ neutrons. This is the common carbon of living matter.
- ${}^{14}_{6}\mathrm{C}$: same element ($Z = 6$), but $N = 8$ neutrons. It is an isotope of carbon — the famous "carbon-14" used for dating archaeological remains.
- ${}^{235}_{92}\mathrm{U}$ and ${}^{238}_{92}\mathrm{U}$: two isotopes of uranium, with 143 and 146 neutrons respectively. Only the first undergoes fission easily in a reactor.

Two neighbouring notions complete the vocabulary:

DEFINITION **Isobars and isotones**

Isobars are nuclides with the same mass number A but different Z (e.g. ${}^{14}_{6}\mathrm{C}$ and ${}^{14}_{7}\mathrm{N}$).
Isotones are nuclides with the same number of neutrons N but different Z (e.g. ${}^{13}_{6}\mathrm{C}$ and ${}^{14}_{7}\mathrm{N}$, both with $N = 7$).

WARNING **Do not confuse the numbers!**

- A is a count of nucleons (a whole number, no unit). It is not the atomic mass in u: the mass of ${}^{12}_{6}\mathrm{C}$ is 12.000 u, but that of ${}^{14}_{6}\mathrm{C}$ is 14.003 u, not exactly 14.
- Z is the number of protons, not the number of neutrons. The neutron number is $N = A - Z$.

- The chemical element is fixed by Z alone: changing N gives an isotope of the same element; changing Z changes the element itself.

1.2 Nuclear Size, Density and the Strong Nuclear Force

How big is a nucleus? Scattering experiments (firing alpha particles or electrons at nuclei, in the spirit of Rutherford) show that the nuclear radius grows with the mass number according to

$$R = r_0 A^{1/3}, \qquad r_0 \approx 1.2 \times 10^{-15}\ \text{m} = 1.2\ \text{fm}.$$

The unit $1\ \text{fm} = 10^{-15}$ m is called the femtometre (or "fermi"). For $^{238}_{92}\text{U}$, $R \approx 1.2 \times 10^{-15} \times 238^{1/3} \approx 7.4 \times 10^{-15}$ m.

Since the volume $V = \frac{4}{3}\pi R^3 = \frac{4}{3}\pi r_0^3 A$ is proportional to A, and the mass is also proportional to A, the **density of nuclear matter is the same for all nuclei**:

$$\rho = \frac{A\,m_n}{\frac{4}{3}\pi r_0^3 A} = \frac{3m_n}{4\pi r_0^3} \approx \frac{3 \times 1.675 \times 10^{-27}}{4\pi (1.2 \times 10^{-15})^3} \approx 2.3 \times 10^{17}\ \text{kg m}^{-3}.$$

One teaspoon of pure nuclear matter would weigh hundreds of millions of tonnes. Nuclei are the densest ordinary objects in the universe.

A paradox. Here is the puzzle that should bother every thoughtful student. The nucleus contains several protons packed into a ball a few femtometres wide. But protons all carry charge $+e$, and Coulomb's law says like charges repel with a force

$$F_{\text{elec}} = \frac{e^2}{4\pi\varepsilon_0 r^2} \approx \frac{9 \times 10^9 \times (1.6 \times 10^{-19})^2}{(2 \times 10^{-15})^2} \approx 58\ \text{N},$$

an enormous force on a particle of mass 10^{-27} kg! Why does the nucleus not fly apart instantly? There must exist a third force of nature, stronger than electricity at these distances: the **strong nuclear force**.

PROPERTY **The strong nuclear force**

The strong nuclear force acts between nucleons (proton–proton, proton–neutron, neutron–neutron alike) and has the following features:

1. Attractive and about 100 times stronger than the electrostatic repulsion at distances of order 1 fm;
2. Very short range: it is essentially zero beyond about 2 to 3 fm;
3. Charge-independent: it does not distinguish protons from neutrons;
4. Saturating: a nucleon binds strongly only to its few nearest neighbours.

(The precise mathematical form of this force is not required at GCE Advanced Level; the four properties above are examinable.)

1.3 The Origin of Nuclear Instability: a Competition of Forces

The stability of a nucleus is the outcome of a competition:

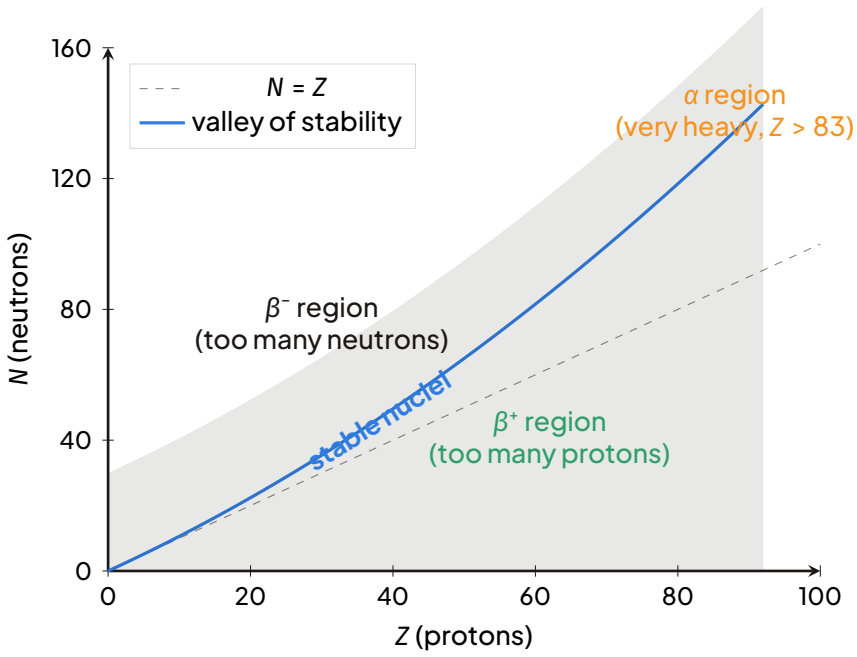
- the **strong force** attracts every nucleon towards its immediate neighbours and holds the nucleus together;
- the **electrostatic force** makes every proton repel every other proton (Coulomb's force has infinite range), and pushes the nucleus apart.

Because the strong force saturates (each nucleon feels only its neighbours) while the Coulomb repulsion grows with the total number of proton pairs, the repulsion wins more and more as Z increases. This single fact explains the whole architecture of nuclear stability:

- **Light nuclei** ($Z \lesssim 20$): a stable nucleus has roughly equal numbers of protons and neutrons, $N \approx Z$. Examples: ${}^{4}_{2}\mathrm{He}$, ${}^{12}_{6}\mathrm{C}$, ${}^{16}_{8}\mathrm{O}$, ${}^{40}_{20}\mathrm{Ca}$.
- **Heavy nuclei**: extra neutrons are needed. Neutrons add strong-force attraction without adding any Coulomb repulsion — they act like a nuclear "cement" diluting the proton repulsion. Stable heavy nuclei therefore have $N > Z$: for ${}^{208}_{82}\mathrm{Pb}$, $N = 126$ against $Z = 82$, i.e. $N/Z \approx 1.5$.
- **Beyond** $Z = 83$: no amount of neutron cement can compensate the repulsion of so many protons. All nuclei with $Z > 83$ are unstable (radioactive), including uranium.

1.4 The Valley of Stability

Plotting the neutron number N against the proton number Z for all known nuclides gives one of the most instructive charts in physics. The stable nuclides cluster along a narrow band, the **valley of stability**, which starts along the line $N = Z$ and then curves towards $N > Z$.



The N–Z chart. Stable nuclides lie in the valley of stability (blue curve). Nuclides above the valley are neutron-rich and decay by β^- emission; those below are proton-rich and decay by β^+ emission; very heavy nuclides ($Z > 83$) decay mainly by α emission.

The chart tells us, at a glance, how an unstable nucleus will try to reach stability:

- A nucleus **above** the valley has too many neutrons. It converts a neutron into a proton by **beta-minus** (β^-) emission, moving one step down towards the valley.
- A nucleus **below** the valley has too many protons. It converts a proton into a neutron by **beta-plus** (β^+) emission (or electron capture).
- A **very heavy** nucleus ($Z > 83$) sheds bulk by ejecting a ${}^4_2\mathrm{He}$ nucleus: **alpha** (α) emission.

1.5 Radioactivity and Soddy's Conservation Laws

DEFINITION **Radioactivity**

Radioactivity is the spontaneous disintegration (transformation) of an unstable nucleus into a different, more stable nucleus, accompanied by the emission of radiation (α particles, β particles and/or γ photons). It is a nuclear phenomenon: it depends only on the nucleus, and is unaffected by temperature, pressure, chemical combination or electric fields.

In a nuclear transformation, the particles rearrange, but two bookkeeping rules always hold. They were stated by Frederick Soddy and Kazimierz Fajans around 1913.

PROPERTY **Soddy's conservation laws**

In every nuclear equation, the following are conserved:

1. the total mass number A: the sum of the superscripts is the same on both sides (conservation of the number of nucleons);
2. the total charge number Z: the sum of the subscripts is the same on both sides (conservation of electric charge).

For a reaction ${}^{A_1}_{Z_1}X + {}^{A_2}_{Z_2}Y \rightarrow {}^{A_3}_{Z_3}X' + {}^{A_4}_{Z_4}Y'$:

$$A_1 + A_2 = A_3 + A_4 \qquad \text{and} \qquad Z_1 + Z_2 = Z_3 + Z_4.$$

(The proof is examinable in the trivial sense: you must be able to apply and justify both equalities in any nuclear equation.)

Note carefully: Soddy's laws conserve the number of nucleons, not the mass. As we shall see in Section 3, the total mass is not conserved — the small "mass defect" is exactly what releases nuclear energy.

METHOD **Checking (or completing) a nuclear equation**

1. Write the equation with the symbols ${}^{A}_{Z}X$ for every species, including particles: $\alpha = {}^{4}_{2}He$, $\beta^- = {}^{0}_{-1}e$, $\beta^+ = {}^{0}_{+1}e$, neutron = ${}^{1}_{0}n$, proton = ${}^{1}_{1}p$, $\gamma = {}^{0}_{0}\gamma$.
2. Balance A: sum the superscripts on the left; the sum of the superscripts on the right must be equal. Solve for any unknown A.
3. Balance Z: do the same with the subscripts. Solve for any unknown Z.
4. Identify the element from Z using the periodic table (Z alone fixes the chemical symbol).
5. Re-read the final equation and double-check both sums.

WORKED EXAMPLE 1 **Completing a decay equation**

Radium-226, once used in hospital sources, decays by alpha emission into radon (Rn, $Z = 86$). Write the balanced equation.

SOLUTION

Step 1 — skeleton equation. The alpha particle is ${}^{4}_{2}He$:

$${}^{226}_{88}\text{Ra} \longrightarrow {}^{A}_{Z}\text{Rn} + {}^{4}_{2}\text{He}.$$

Step 2 — conservation of A:

$$226 = A + 4 \quad \Rightarrow \quad A = 222.$$

Step 3 — conservation of Z:

$$88 = Z + 2 \quad \Rightarrow \quad Z = 86.$$

Step 4 — identification. $Z = 86$ is indeed radon, consistent with the statement. Hence

$$\boxed{{}^{226}_{88}\mathrm{Ra} \longrightarrow {}^{222}_{86}\mathrm{Rn} + {}^{4}_{2}\mathrm{He}}$$

Check: $226 = 222 + 4$ ✓ and $88 = 86 + 2$ ✓. The daughter nucleus has $N = 222 - 86 = 136$ neutrons and lies closer to the valley of stability than radium-226.

WORKED EXAMPLE 2 **Identifying an unknown particle**

When nitrogen-14 in the upper atmosphere is struck by a neutron, carbon-14 is produced together with a second particle. Identify it.

SOLUTION

Step 1: ${}^{14}_{7}\mathrm{N} + {}^{1}_{0}\mathrm{n} \longrightarrow {}^{14}_{6}\mathrm{C} + {}^{A}_{Z}\mathrm{X}$.
Step 2 (balance A): $14 + 1 = 14 + A \Rightarrow A = 1$.
Step 3 (balance Z): $7 + 0 = 6 + Z \Rightarrow Z = 1$.
Step 4: ${}^{1}_{1}X$ is a proton, ${}^{1}_{1}p$ (a hydrogen nucleus). The full equation is

$$ {}^{14}_{7}\mathrm{N} + {}^{1}_{0}\mathrm{n} \longrightarrow {}^{14}_{6}\mathrm{C} + {}^{1}_{1}\mathrm{p}.$$

This is precisely how the carbon-14 used in archaeological dating is continuously created in the atmosphere by cosmic rays.

WARNING **Frequent examination traps**

- Writing β^- as ${}^{0}_{1}e$: the electron has charge $-e$, so its subscript is −1, not +1. With ${}^{0}_{-1}e$, the charge balance works automatically.
- Believing that A is conserved "because mass is conserved": in nuclear physics mass is not conserved (Section 3). What is conserved is the number of nucleons and the charge.
- Forgetting the emitted particle altogether when balancing: always check whether an α, β or γ appears on the right-hand side.
- Confusing the daughter element: after balancing Z, look up the symbol — never keep

the parent's symbol with a new Z.

- Ignoring γ emission: a γ photon (${}^{0}_{0}\gamma$) carries away energy but changes neither A nor Z. It often accompanies α or β decay but must be included if the question mentions it.

DID YOU KNOW? **Radioactivity all around us**

Radioactivity is not an exotic laboratory curiosity. The potassium-40 in bananas and in the volcanic soils of the Mount Cameroon region, the radon gas seeping from granitic rocks, and the cosmic rays that generate carbon-14 high above the Sahel all make natural radioactivity part of everyday life. The same carbon-14, absorbed by plants and then by animals, allows scientists to date fossils and archaeological remains up to about 50 000 years old — a direct application of the nuclear physics of this chapter.

KEY POINTS **Key points of this section**

- The nucleus contains Z protons and $N = A - Z$ neutrons, written ${}^{A}_{Z}X$; nucleons = protons + neutrons.
- Isotopes: same Z, different N (e.g. ${}^{12}_{6}C$ / ${}^{14}_{6}C$, ${}^{235}_{92}U$ / ${}^{238}_{92}U$). Isobars: same A. Isotones: same N.
- Nuclear radius $R = r_0 A^{1/3}$ with $r_0 \approx 1.2$ fm; nuclear density $\approx 2.3 \times 10^{17}$ kg m^{-3}, the same for all nuclei.
- The strong nuclear force is attractive, very short-ranged (~ 1 fm), charge-independent, and about 100 times stronger than Coulomb repulsion at contact.
- Stability results from the strong-versus-Coulomb competition: $N \approx Z$ for light nuclei, $N > Z$ for heavy ones; no stable nucleus exists for $Z > 83$.
- On the N–Z chart, unstable nuclides lie off the valley of stability: β^- above, β^+ below, α for the very heavy.
- Radioactivity is the spontaneous disintegration of an unstable nucleus with emission of radiation.
- Soddy's laws: in every nuclear equation, $\sum A$ and $\sum Z$ are conserved.

EXERCISE 1 **Practice**

1. Give the number of protons, neutrons and electrons of the neutral atom $^{60}_{27}\mathrm{Co}$ (a source used in radiotherapy).
2. Balance the following equations and name the emitted particle:
   (a) $^{14}_{6}\mathrm{C} \rightarrow ^{14}_{7}\mathrm{N} + \ldots$
   (b) $^{238}_{92}\mathrm{U} \rightarrow ^{234}_{90}\mathrm{Th} + \ldots$
3. Explain, in terms of the two competing forces, why stable heavy nuclei need $N > Z$.

SOLUTION **Exercise 1**

1. $Z = 27$ protons; $N = A - Z = 60 - 27 = 33$ neutrons; the neutral atom has 27 electrons.
2. (a) Balance A: $14 = 14 + 0$; balance Z: $6 = 7 + Z \Rightarrow Z = -1$. The particle is $^{0}_{-1}e$: a β^- emission, $^{14}_{6}\mathrm{C} \longrightarrow ^{14}_{7}\mathrm{N} + ^{0}_{-1}\mathrm{e}$.
   (b) Balance A: $238 = 234 + 4$; balance Z: $92 = 90 + 2$. The particle is $^{4}_{2}He$: an α emission, $^{238}_{92}\mathrm{U} \longrightarrow ^{234}_{90}\mathrm{Th} + ^{4}_{2}\mathrm{He}$.
3. Each proton repels all other protons (long-range Coulomb force), while the strong force binds only nearest neighbours. As Z grows, extra neutrons add strong attraction without adding repulsion, so a larger neutron excess $N > Z$ is needed to hold the nucleus together.

2 TYPES OF RADIATION, PROPERTIES AND APPLICATIONS

In the previous section we saw that some nuclei are **unstable**: their proton-to-neutron ratio lies outside the band of stability, or they are simply too heavy for the strong nuclear force to hold them together. Such nuclei do not remain unstable forever. Sooner or later, each one spontaneously transforms itself into a more stable configuration by emitting particles or energy. This spontaneous transformation is called **radioactivity**, and the emitted particles or photons are called **radiations**. In this section we study the nature of these radiations, their properties, how we detect them, how we protect ourselves from them, and how humanity has learned to put them to work.

DEFINITION **Radioactivity**

Radioactivity is the spontaneous and random transformation of an unstable nucleus into

a more stable nucleus, accompanied by the emission of radiation (α, β or γ). The original nucleus is called the **parent nucleus** and the nucleus produced is the **daughter nucleus**. The process is:

- spontaneous: it cannot be triggered, accelerated or slowed down by physical or chemical means (temperature, pressure, catalysts have no effect);
- random: it is impossible to predict which nucleus will decay or when; only statistical laws apply to a large sample;
- inevitable: an unstable nucleus will always decay eventually.

Every nuclear decay obeys two conservation rules, known as **Soddy's laws** (or conservation laws of nuclear reactions):

PROPERTY **Soddy's conservation laws**

In any nuclear transformation:

1. the total nucleon number A is conserved: $\sum A_{\text{before}} = \sum A_{\text{after}}$;
2. the total charge number Z is conserved: $\sum Z_{\text{before}} = \sum Z_{\text{after}}$.

These two laws are sufficient to complete any decay equation. The proof is not examinable, but their use is required at the GCE.

2.1 Alpha Radiation

Observation. When a sample of uranium salt is placed near a photographic plate wrapped in thick black paper, the plate is still darkened. But a single sheet of ordinary paper placed between the source and the plate is enough to stop the effect almost completely. The radiation responsible is easily absorbed and strongly ionising: this is **alpha radiation**.

DEFINITION **Alpha radiation**

An **alpha particle** (α) is a helium-4 nucleus ${}^{4}_{2}\text{He}$, i.e. a cluster of 2 protons and 2 neutrons. Its characteristics are:

- mass: $m_\alpha \approx 4\ \text{u}$ (about 6.64×10^{-27} kg);
- charge: $q_\alpha = +2e = +3.20 \times 10^{-19}$ C;
- typical kinetic energy: a few MeV, giving speeds of order $10^7\ \text{m s}^{-1}$ (non-relativistic).

Alpha decay occurs in very **heavy** nuclei ($Z > 82$ roughly), which are unstable simply because they contain too many nucleons. By ejecting an α particle, the nucleus loses 4 nucleons and 2

units of charge and moves closer to stability:

$$^{A}_{Z}X \longrightarrow {}^{A-4}_{Z-2}Y + {}^{4}_{2}He$$

EXEMPLE **Alpha decay of uranium-238**

$$^{238}_{92}U \longrightarrow {}^{234}_{90}Th + {}^{4}_{2}He$$

Check Soddy's laws: nucleons 238 = 234 + 4 ✓; charge 92 = 90 + 2 ✓. Uranium-238 is the most abundant isotope of natural uranium, mined in several African countries.

2.2 Beta-Minus Radiation

Observation. Some radiations pass easily through paper but are stopped by a few millimetres of aluminium. They are deflected by a magnetic field in the opposite direction to alpha particles, and much more strongly. These are **beta-minus particles**.

DEFINITION **Beta-minus radiation**

A **beta-minus particle** (β^-) is an electron $^{0}_{-1}e$ emitted by the nucleus. It has:

- mass: $m_e = 9.11 \times 10^{-31}$ kg (about 1/7300 of an α particle);
- charge: $q = -e = -1.60 \times 10^{-19}$ C;
- speed: up to about 0.9 c (relativistic speeds are common).

The electron is not one of the atom's orbital electrons: it is created inside the nucleus when a neutron transforms into a proton, emitting also an antineutrino $\bar{\nu}_e$ (often omitted in GCE equations but required for energy-momentum conservation):

$$^{1}_{0}n \rightarrow {}^{1}_{1}p + {}^{0}_{-1}e + \bar{\nu}_e$$

Beta-minus decay occurs in nuclei with an **excess of neutrons** (below the band of stability). Converting a neutron into a proton increases Z by 1 while A stays constant:

$$^{A}_{Z}X \longrightarrow {}^{A}_{Z+1}Y + {}^{0}_{-1}e$$

EXEMPLE **Beta-minus decay of carbon-14**

$$^{14}_{6}C \longrightarrow {}^{14}_{7}N + {}^{0}_{-1}e$$

Check: 14 = 14 + 0 ✓; 6 = 7 - 1 ✓. This decay is the basis of carbon-14 dating, discussed below.

2.3 Beta-Plus Radiation

Nuclei with an **excess of protons** (above the band of stability) can convert a proton into a neutron by emitting a **positron** ${}^{0}_{+1}\mathrm{e}$ — the antiparticle of the electron, with the same mass but charge $+e$, together with a **neutrino** ν_e:

$$ {}^{1}_{1}\mathrm{p} \rightarrow {}^{1}_{0}\mathrm{n} + {}^{0}_{+1}\mathrm{e} + \nu_e \qquad \text{so} \qquad {}^{A}_{Z}\mathrm{X} \longrightarrow {}^{A}_{Z-1}\mathrm{Y} + {}^{0}_{+1}\mathrm{e} $$

Example: ${}^{22}_{11}\mathrm{Na} \longrightarrow {}^{22}_{10}\mathrm{Ne} + {}^{0}_{+1}\mathrm{e}$. Positron emitters are used in medicine (PET scanners). At GCE level you must recognise the equation and apply Soddy's laws.

2.4 Gamma Radiation

Observation. After an α or β decay, the daughter nucleus is often left in an **excited state** (denoted *), like an atom whose electrons have been raised to a higher level. It gets rid of this surplus energy by emitting a very penetrating radiation that is not deflected by any field: **gamma radiation**.

DEFINITION **Gamma radiation**

A **gamma ray** (γ) is a high-energy electromagnetic photon, of the same nature as light and X-rays but far more energetic (frequencies above 10^{19} Hz, energies typically 0.1 to several MeV). It has:

- no mass and no charge;
- speed $c = 3.00 \times 10^{8}\ \mathrm{m\,s^{-1}}$;
- energy $E = hf$.

Gamma emission is the de-excitation of a nucleus:

$$ {}^{A}_{Z}\mathrm{X}^{*} \rightarrow {}^{A}_{Z}\mathrm{X} + \gamma $$

Neither A nor Z changes: the nucleus keeps its identity, it only loses energy. Gamma emission usually accompanies an α or β decay.

WARNING **Common mistakes to avoid**

- Gamma decay does NOT change A or Z. Writing ${}^{A}_{Z}\mathrm{X} \rightarrow {}^{A-4}_{Z-2}\mathrm{Y} + \gamma$ is doubly wrong. Only α and β decays change the nucleus; γ emission is pure energy release.
- The alpha particle is NOT an electron. It is a helium nucleus ${}^{4}_{2}\mathrm{He}$, about 7300 times heavier than an electron and positively charged. Confusing α with β^{-} in an equation destroys both conservation laws.
- In β^{-} decay, A stays constant and Z increases by 1 — many candidates mistakenly decrease it.

- In β^+ decay, A stays constant and Z decreases by 1.

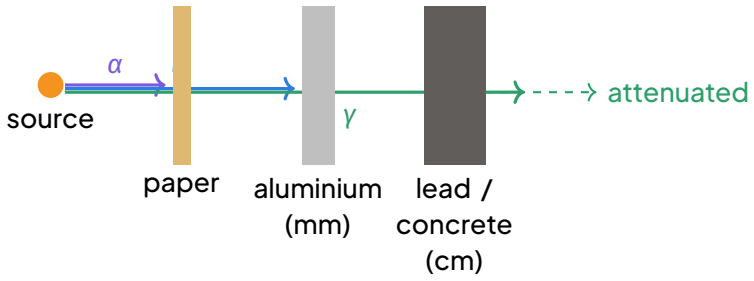
2.5 Comparative Properties of the Radiations

Because of their different masses and charges, the three radiations interact very differently with matter. A heavy, doubly-charged α particle ploughs through atoms, ripping electrons off (ionising) at every step, and therefore loses its energy very quickly: it is **strongly ionising but weakly penetrating**. A light β particle ionises less and travels further. A neutral γ photon interacts only occasionally, so it ionises weakly but penetrates deeply.

PROPERTY **Comparative properties of α, β, γ radiations**

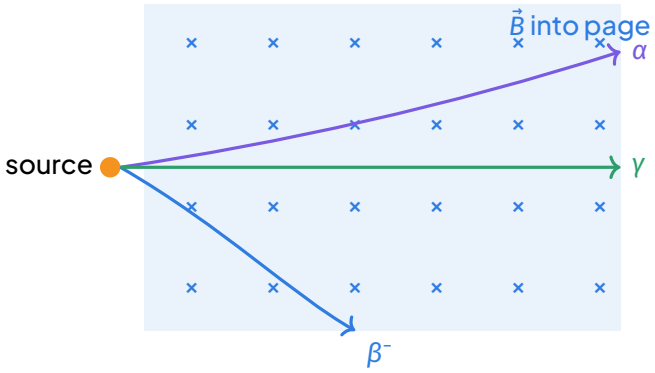
Property	Alpha α	Beta β^-	Gamma γ
Nature	$^4_2$He nucleus	electron $^0_{-1}$e	EM photon
Mass	≈ 4 u	9.11×10^{-31} kg	0
Charge	$+2e$	$-e$	0
Ionising power	very strong (≈ 10^4 ion pairs per cm in air)	moderate (≈ 10^2)	weak (≈ 1)
Penetrating power	weak	moderate	very strong
Range in air	a few cm	a few m	hundreds of m
Stopped by	sheet of paper, skin	a few mm of aluminium	thick lead or concrete (only attenuated)
In $\vec{E}$ or $\vec{B}$ field	deflected, slightly	deflected strongly, opposite way	not deflected

Memory aid: ionising power $\alpha > \beta > \gamma$; penetrating power $\gamma > \beta > \alpha$. The two orders are reversed: the better a radiation ionises, the faster it is stopped.



Penetrating power: α is stopped by paper, β by a few millimetres of aluminium, while γ is only attenuated by thick lead or concrete.

In a uniform magnetic field $\vec{B}$ directed into the page, the magnetic force $\vec{F} = q\,\vec{v} \wedge \vec{B}$ curves the trajectory of charged particles into a circular arc of radius $R = \dfrac{mv}{|q|B}$. The α particle (heavy, charge $+2e$) curves slightly one way; the β^- (light, charge $-e$) curves sharply the opposite way; the γ photon goes straight through.



Paths of α, β^- and γ radiations in a uniform magnetic field. The curvature radius $R = mv/(|q|B)$ is large for the heavy α, small for the light β^-, and infinite for the neutral γ.

METHOD **Predicting the daughter nucleus with Soddy's laws**

To complete a decay equation ${}^{A}_{Z}\mathrm{X} \rightarrow {}^{A'}_{Z'}\mathrm{Y}$ + radiation:

1. Identify the radiation and write its symbol with its numbers: $\alpha = {}^{4}_{2}\mathrm{He}$, $\beta^- = {}^{0}_{-1}\mathrm{e}$, $\beta^+ = {}^{0}_{+1}\mathrm{e}$, $\gamma = {}^{0}_{0}\equiv$.
2. Conserve nucleons: $A' = A - A_{\text{radiation}}$.
3. Conserve charge: $Z' = Z - Z_{\text{radiation}}$ (careful with the negative charge of β^-: $Z' = Z - (-1) = Z + 1$).
4. Identify the element from Z' using the periodic table, and verify both conservation laws on the final equation.

WORKED EXAMPLE 3 **Completing decay equations (GCE style)**

Statement. (a) Radium-226 $^{226}_{88}\mathrm{Ra}$ is an α emitter. Write its decay equation, given that the daughter is radon (Rn). (b) Cobalt-60 $^{60}_{27}\mathrm{Co}$ emits a β^- particle and then a γ photon. Write both equations (daughter: nickel, Ni).

SOLUTION

Solution. (a) Alpha emission removes 4 nucleons and 2 charges:

$$^{226}_{88}\mathrm{Ra} \longrightarrow {}^{226-4}_{88-2}\mathrm{Rn} + {}^{4}_{2}\mathrm{He} \quad \Longrightarrow \quad {}^{226}_{88}\mathrm{Ra} \longrightarrow {}^{222}_{86}\mathrm{Rn} + {}^{4}_{2}\mathrm{He}$$

Check: 226 = 222 + 4 ✓; 88 = 86 + 2 ✓.
(b) Beta-minus emission keeps A constant and increases Z by 1:

$$^{60}_{27}\mathrm{Co} \longrightarrow {}^{60}_{28}\mathrm{Ni}^{*} + {}^{0}_{-1}\mathrm{e}$$

The nickel nucleus is left in an excited state and de-excites:

$$^{60}_{28}\mathrm{Ni}^{*} \rightarrow {}^{60}_{28}\mathrm{Ni} + \Xi$$

Check: 60 = 60 + 0 ✓; 27 = 28 − 1 ✓. This is exactly the decay used in cobalt-60 radiotherapy.

2.6 Detection of Radiations

Radiations are invisible, but their **ionising power** betrays them. Three detectors you must know qualitatively:

- **Geiger–Müller (GM) counter.** The radiation enters a gas-filled tube through a thin window (mica for α) and ionises the gas; each ionisation triggers a brief electrical discharge which is counted (audible "clicks" or a digital count rate in counts per second). It detects β and γ readily; α only if the window is very thin.
- **Cloud chamber (Wilson chamber).** Air saturated with alcohol vapour is cooled so the vapour is supersaturated. Ions created along a particle's path act as condensation nuclei: a visible track of droplets appears. Alpha particles leave short, thick, straight tracks; β particles leave long, thin, wandering tracks; γ rays leave almost no track (they are detected only through secondary electrons).
- **Photographic film.** Ionising radiation darkens a photographic emulsion exactly like light. Film badges worn by radiographers and hospital staff measure the accumulated dose over weeks.

2.7 Biological Effects and Radiation Protection

When radiation ionises the molecules of living cells, it can damage DNA, kill cells or trigger cancers. The danger depends on the radiation: **outside** the body, α particles are stopped by the dead outer layer of skin and are almost harmless, while γ rays cross the whole body. **Inside** the

body (inhaled or swallowed source), the situation reverses: the intensely ionising α particles deposit all their energy in a tiny volume of tissue and become the most dangerous.

KEY POINTS **The three rules of radiation protection**

1. Distance: keep as far as possible from the source — the received dose decreases rapidly with distance (roughly as $1/d^2$ for a point source). Laboratory sources are handled with tongs, never with fingers.
2. Shielding: place an absorber between source and person — paper or clothing for α, aluminium or plastic for β, thick lead or concrete for γ.
3. Time: minimise the exposure time; the dose received is proportional to the duration of exposure.

2.8 Applications of Radioactivity

Far from being only a hazard, radioactivity is a powerful tool when used in a controlled way.

Medicine.

- **Radiotherapy**: the penetrating γ rays of cobalt-60 are aimed at tumours to destroy cancerous cells (cancer treatment units in reference hospitals, e.g. in Yaoundé and Douala).
- **Radioactive tracers**: a small amount of a short-lived emitter (e.g. iodine-131 for the thyroid, technetium-99m for imaging) is introduced into the body; its radiation, detected from outside, reveals how an organ functions.
- **Sterilisation**: γ rays from cobalt-60 sterilise surgical instruments and syringes sealed in their packaging — no heat, no chemicals.

Agriculture and food.

- **Food irradiation**: γ rays kill bacteria, moulds and insects in stored food (grains, dried fish, spices), greatly extending shelf life without cooking the food.
- **Pest control**: male insects sterilised by irradiation are released; mating with them produces no offspring, reducing the pest population.
- Tracers also follow the uptake of fertilisers by crops, helping optimise their use.

Industry.

- **Thickness gauging**: a β source on one side of a rolling mill and a detector on the other; any change in the thickness of a metal or paper sheet changes the count rate, and the rollers are adjusted automatically.
- **Leak detection**: a radioactive tracer injected into a pipeline reveals leaks by appearing at the surface above the fault.

Carbon-14 dating. Living organisms continually exchange carbon with the atmosphere, so the proportion of radioactive ^{14}C in their tissues stays constant while they live. At death, the ex-

change stops and the ^{14}C content decreases by β^- decay with a half-life $T_{1/2} \approx 5730$ years. Measuring the remaining activity of a wooden tool, a bone or a piece of charcoal and comparing it with that of living matter gives the time elapsed since death:

$$N = N_0 \left(\frac{1}{2}\right)^{t/T_{1/2}}$$

This links radioactivity directly to the law of radioactive decay and half-life studied elsewhere in this chapter — a favourite GCE combination.

DID YOU KNOW? **Did you know?**

Radioactivity was discovered in 1896 by Henri Becquerel, who noticed that uranium salts fogged photographic plates kept in a dark drawer. Marie and Pierre Curie then isolated two new radioactive elements, polonium and radium. Rutherford later separated the emissions into "α" and "β" rays using a magnetic field, and Villard identified the undeflected "γ" rays — the very classification you have just studied.

WARNING **Exam tip — what the GCE Board loves to ask**

GCE Paper 2 questions very frequently ask you to:

- complete a decay equation (apply Soddy's laws and show the check — it earns marks);
- identify a radiation from its properties: "stopped by paper" ⇒ α; "deflected opposite to α and more strongly" ⇒ β^-; "undeflected, needs lead shielding" ⇒ γ;
- state one application and one hazard of a given radioisotope.

Learn the comparative table by heart: it answers most of these questions in one line.

EXERCISE 2 **Practice**

1. Polonium-210 $^{210}_{84}Po$ decays by α emission into lead (Pb). Write the equation.
2. A radiation passes through paper but is stopped by 3 mm of aluminium and is deflected towards the positive plate of an electric field. Identify it and justify.
3. Explain why an α source is safe outside the body but dangerous if swallowed.
4. State one medical and one industrial application of radioactivity, naming the radiation used in each case.

SOLUTION **Exercise 1**

1. $^{210}_{84}\mathrm{Po} \longrightarrow {}^{206}_{82}\mathrm{Pb} + {}^{4}_{2}\mathrm{He}$. Check: 210 = 206 + 4; 84 = 82 + 2.
2. It is β^- radiation: it penetrates paper but not a few mm of aluminium (moderate penetrating power), and its deflection towards the positive plate shows a negative charge.
3. Outside the body, α particles are stopped by the dead outer layer of skin before reaching living cells. If the source is swallowed or inhaled, the α particles are emitted directly inside living tissue; their very high ionising power then destroys cells in a small volume, causing severe damage.
4. Medical: cobalt-60 radiotherapy, using penetrating γ rays to destroy tumours (or iodine-131 tracer, β^-/γ). Industrial: thickness gauging of metal sheets using a β source and detector (or pipeline leak detection with a tracer).

3 MASS DEFECT AND NUCLEAR ENERGY

In the previous section we studied the radiations emitted by unstable nuclei and their applications. A natural question now arises: where does the energy carried away by these radiations come from? The answer lies in one of the most profound discoveries of the twentieth century: mass and energy are two faces of the same reality. By carefully weighing nuclei, physicists discovered that a bound nucleus is slightly lighter than its separated nucleons, and that the "missing mass" corresponds exactly to the energy that holds the nucleus together. This is the key to understanding radioactivity, nuclear power and the energy of the stars.

3.1 The Atomic Mass Unit and the Masses of the Nucleons

Nuclear masses are extremely small in kilograms, so physicists use a more convenient unit.

DEFINITION **Atomic mass unit**

The **atomic mass unit**, symbol u, is defined as one twelfth of the mass of a neutral atom of carbon-12, $^{12}_{6}\mathrm{C}$:

$$1\,\mathrm{u} = \frac{m(^{12}_{6}\mathrm{C})}{12} = 1.66054 \times 10^{-27}\ \mathrm{kg}.$$

The masses of the three particles we shall need are:

Particle	Mass in u	Mass in kg
Proton	$m_p = 1.00728\,\mathrm{u}$	$1.6726 \times 10^{-27}\ \mathrm{kg}$
Neutron	$m_n = 1.00866\,\mathrm{u}$	$1.6749 \times 10^{-27}\ \mathrm{kg}$
Electron	$m_e = 0.00055\,\mathrm{u}$	$9.11 \times 10^{-31}\ \mathrm{kg}$

Note that the neutron is slightly heavier than the proton: this small difference has important

consequences (a free neutron is unstable and undergoes β^- decay into a proton, an electron and an antineutrino).

3.2 Observation: The Nucleus Weighs Less Than Its Parts

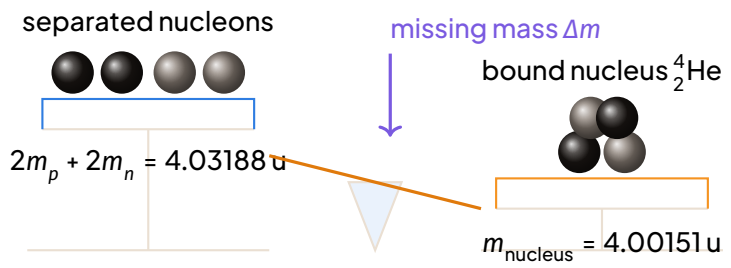
Consider the helium-4 nucleus, ${}^4_2\mathrm{He}$ (the α particle). It contains $Z = 2$ protons and $A - Z = 2$ neutrons. Let us add the masses of the separated nucleons:

$$2m_p + 2m_n = 2(1.00728) + 2(1.00866) = 4.03188\ \mathrm{u}.$$

Now the measured mass of the helium-4 nucleus is $m_{\text{nucleus}} = 4.00151\ \mathrm{u}$. The bound nucleus is lighter by about 0.03037 u! This is a general experimental fact:

WARNING **A surprising but universal observation**

For every stable nucleus, the mass of the nucleus is strictly less than the sum of the masses of its separated nucleons. Mass has "disappeared" during the formation of the nucleus. As we shall see, it has in fact been converted into energy and carried away.



The mass balance: four separated nucleons (2 protons + 2 neutrons) are heavier than the bound helium-4 nucleus. The difference Δm is the mass defect.

3.3 Mass Defect: Definition

DEFINITION **Mass defect**

The **mass defect** Δm of a nucleus ${}^A_Z\mathrm{X}$ is the difference between the total mass of its Z protons and $(A - Z)$ neutrons taken separately, and the measured mass of the nucleus:

$$\Delta m = \left[Z\,m_p + (A - Z)\,m_n\right] - m_{\text{nucleus}}$$

The mass defect is always positive for a bound nucleus. It is expressed in u or in kg.

WARNING **Nucleus mass or atom mass?**

Data tables often give the mass of the neutral atom (nucleus + electrons), not of the bare nucleus. If you use atomic masses, you must use $Z\,m({}^{1}_{1}\mathrm{H})$ (hydrogen atom = proton + electron) instead of $Z\,m_p$, so that the electron masses cancel. Mixing atomic masses with proton masses is a classic source of error at the GCE.

3.4 Einstein's Mass–Energy Equivalence

In 1905, Einstein showed from the theory of special relativity that mass is a form of energy.

PROPERTY **Einstein's mass–energy relation**

Any body of mass m possesses, simply because of its mass, an energy

$$\boxed{E = mc^2}$$

where $c = 3.00 \times 10^8\ \mathrm{m\,s^{-1}}$ is the speed of light in vacuum. Conversely, whenever a system releases energy ΔE, its mass decreases by $\Delta m = \Delta E / c^2$.

Because $c^2 \approx 9 \times 10^{16}\ \mathrm{m^2\,s^{-2}}$ is enormous, a tiny mass corresponds to a huge energy: 1 kg of matter "contains" 9×10^{16} J, roughly the energy produced by a large power station in several years.

The conversion factor 1 u ↔ 931.5 MeV. Let us compute the energy equivalent of one atomic mass unit, using the precise values $1\ \mathrm{u} = 1.66054 \times 10^{-27}$ kg and $c = 2.998 \times 10^8\ \mathrm{m\,s^{-1}}$:

$$E = (1\ \mathrm{u})\,c^2 = 1.66054 \times 10^{-27} \times (2.998 \times 10^8)^2 = 1.4924 \times 10^{-10}\ \mathrm{J}.$$

Converting to electron-volts ($1\ \mathrm{eV} = 1.60 \times 10^{-19}$ J):

$$E = \frac{1.4924 \times 10^{-10}}{1.602 \times 10^{-19}} = 9.315 \times 10^8\ \mathrm{eV} = 931.5\ \mathrm{MeV}.$$

KEY POINTS **The key conversion**

$$\boxed{1\ \mathrm{u} \times c^2 \approx 931.5\ \mathrm{MeV}}$$

This factor lets you convert a mass defect in u directly into an energy in MeV, without passing through kilograms and joules. Memorise it.

3.5 Binding Energy of the Nucleus

To separate a nucleus into its individual nucleons, one must supply energy to overcome the strong nuclear attraction. Conversely, when the nucleus forms from separated nucleons, that

energy is released (carried away by a photon, for instance), which is exactly why the nucleus ends up lighter.

DEFINITION **Binding energy**

The **binding energy** E_l of a nucleus is the energy that must be supplied to separate it completely into its constituent nucleons at rest. It equals the energy equivalent of the mass defect:

$$E_l = \Delta m \cdot c^2 = \left[Z m_p + (A - Z) m_n - m_{\text{nucleus}}\right] c^2$$

It is expressed in joules (SI) or, more conveniently, in MeV.

EXEMPLE **Binding energy of helium-4**

For ^4_2He, we found $\Delta m = 4.03188 - 4.00151 = 0.03037$ u. Hence

$$E_l = 0.03037 \times 931.5 = 28.3 \text{ MeV} = 28.3 \times 10^6 \times 1.60 \times 10^{-19} = 4.53 \times 10^{-12} \text{ J}.$$

About 28 MeV is needed to tear a helium nucleus apart: this is why α particles are so stable.

3.6 Binding Energy per Nucleon and the Aston Curve

Is a heavy nucleus more stable than a light one simply because its total binding energy is larger? No: a heavy nucleus also has more nucleons to bind. The fair comparison is the average binding energy of each nucleon.

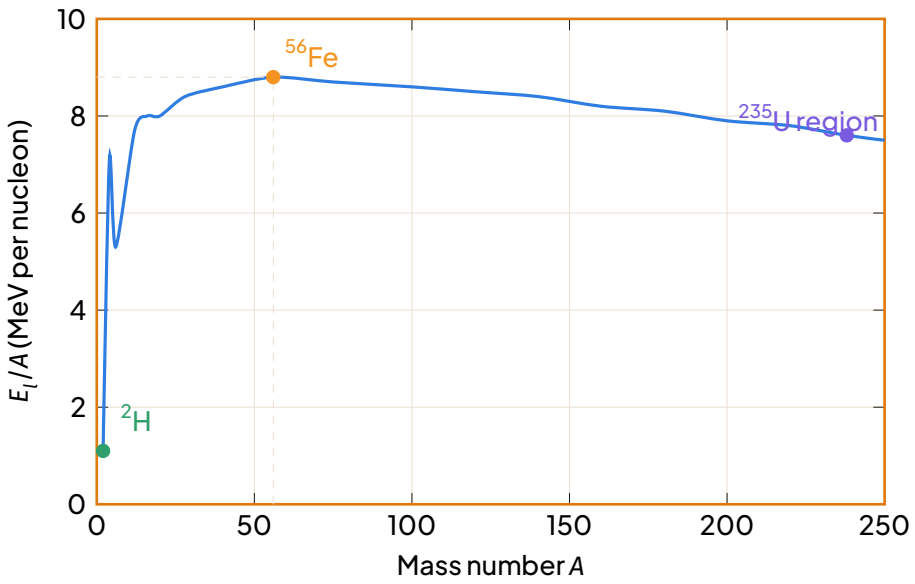
PROPERTY **Binding energy per nucleon measures stability**

The **binding energy per nucleon** is

$$\frac{E_l}{A}$$

The larger E_l/A, the more stable the nucleus, because more energy per nucleon is needed to dismantle it. The maximum of E_l/A (about 8.8 MeV per nucleon) occurs near iron-56, $^{56}_{26}\text{Fe}$, which is therefore one of the most stable nuclei in nature.

The graph of E_l/A against the mass number A is called the **Aston curve**.



The Aston curve: binding energy per nucleon versus mass number. The maximum (≈ 8.8 MeV/nucleon) is near iron-56. Very light nuclei (left) can gain stability by fusion; very heavy nuclei (right) by fission.

Consequences of the shape of the curve:

- Heavy nuclei ($A \gtrsim 200$, e.g. uranium-235) sit on the descending part. If such a nucleus splits into two medium nuclei (fission), the products lie higher on the curve: they are more stable, and the difference in binding energy is released.
- Light nuclei ($A \lesssim 20$, e.g. ^{2}H, ^{3}H) sit on the steep rising part. If two light nuclei merge (fusion), the product is more stable and energy is released. This powers the Sun.
- Medium nuclei near iron are at the top: neither fission nor fusion of iron releases energy. Iron is the "nuclear ash" of stars.

3.7 Energy Balance of a Nuclear Reaction

Any nuclear reaction (radioactive decay, fission, fusion, bombardment) conserves the total mass–energy. If the products are lighter than the reactants, the missing mass has been converted into kinetic energy of the products and into radiation.

PROPERTY **Energy released by a nuclear reaction**

The **energy balance** (or Q-value) of a nuclear reaction is

$$Q = \left(m_{\text{reactants}} - m_{\text{products}}\right) c^2$$

- If $Q > 0$: the reaction is **exoenergetic** (exothermic); it releases energy and can occur spontaneously.

- If $Q < 0$: the reaction is **endoenergetic** (endothermic); it can only occur if the missing energy is supplied, for example as kinetic energy of an incoming projectile.

(The proof is a direct application of conservation of total energy $E = mc^2 + E_k$; it is instructive but not required as a formal proof at the GCE.)

METHOD **Computing the energy released by a nuclear reaction**

1. Write the balanced nuclear equation (conservation of A and Z).
2. Look up the masses (in u) of all reactants and all products.
3. Compute $\Delta m = m_{\text{reactants}} - m_{\text{products}}$ in u.
4. Convert to MeV: $Q\,(\text{MeV}) = \Delta m\,(\text{u}) \times 931.5$.
5. Convert to joules if required: $Q\,(\text{J}) = Q\,(\text{MeV}) \times 10^{6} \times 1.60 \times 10^{-19}$.
6. Conclude: $Q > 0$ means energy is released; $Q < 0$ means energy must be supplied.

WORKED EXAMPLE 4 **Energy released in the α decay of radium-226**

The radium-226 nucleus decays according to

$$^{226}_{88}\text{Ra} \longrightarrow {}^{222}_{86}\text{Rn} + {}^{4}_{2}\text{He}.$$

Given: $m(^{226}_{88}\text{Ra}) = 225.9771$ u; $m(^{222}_{86}\text{Rn}) = 221.9703$ u; $m(^{4}_{2}\text{He}) = 4.0015$ u. Calculate the energy released in MeV and in joules.

SOLUTION

Step 1 – Mass defect of the reaction:

$$\Delta m = m_{\text{reactants}} - m_{\text{products}} = 225.9771 - (221.9703 + 4.0015) = 0.0053\ \text{u}.$$

Step 2 – Conversion to MeV:

$$Q = 0.0053 \times 931.5 = 4.9\ \text{MeV}.$$

Step 3 – Conversion to joules:

$$Q = 4.9 \times 10^{6} \times 1.60 \times 10^{-19} = 7.9 \times 10^{-13}\ \text{J}.$$

Conclusion: $Q > 0$, so the decay is exoenergetic: radium-226 decays spontaneously. The energy appears essentially as the kinetic energy of the α particle (the radon nucleus recoils slightly). Although 7.9×10^{-13} J per decay seems tiny, one gram of radium contains about 2.7×10^{21} nuclei, so the energy released per gram is macroscopic.

WORKED EXAMPLE 5 **Binding energy per nucleon of carbon-12**

The mass of the carbon-12 nucleus is $m_{\text{nucleus}} = 11.99671$ u (obtained by subtracting the six electron masses from the atomic mass, exactly 12 u). Calculate the binding energy per nucleon of ${}^{12}_{6}\text{C}$.

SOLUTION

Step 1 – Mass defect: carbon-12 has $Z = 6$ protons and $A - Z = 6$ neutrons:

$$\Delta m = [6(1.00728) + 6(1.00866)] - 11.99671 = 12.09564 - 11.99671 = 0.09893 \text{ u}.$$

Step 2 – Binding energy:

$$E_l = 0.09893 \times 931.5 = 92.2 \text{ MeV}.$$

Step 3 – Per nucleon:

$$\frac{E_l}{A} = \frac{92.2}{12} = 7.7 \text{ MeV/nucleon}.$$

This is below the iron-56 maximum of 8.8 MeV/nucleon, consistent with the position of carbon on the rising part of the Aston curve.

WARNING **Common mistakes to avoid**

- Forgetting the MeV → J conversion. The SI unit of energy is the joule. If a question asks for the energy in joules, multiply the value in MeV by 1.60×10^{-13} (since $1 \text{ MeV} = 10^{6} \times 1.60 \times 10^{-19}$ J). Leaving the answer in MeV costs marks.
- Mixing atomic and nuclear masses. Either use nuclear masses with m_p everywhere, or use atomic masses with the mass of the hydrogen atom $m({}^{1}_{1}\text{H})$ everywhere. Never mix the two conventions in the same calculation.
- Sign of Δm. The mass defect is reactants minus products. A negative Q does not mean "energy destroyed"; it means the reaction needs an energy input.
- Confusing E_l and E_l/A. A large total binding energy does not imply great stability: uranium has a huge E_l but is unstable because its E_l/A is low. Always compare binding energies per nucleon.
- Rounding too early. Mass defects are tiny differences between large numbers; keep at least four decimal places in the masses (in u) until the final step.

DID YOU KNOW? **Why the Sun shines**

Deep in the Sun's core, four hydrogen nuclei fuse (through a chain of reactions) into one helium-4 nucleus. The helium nucleus is lighter than the four protons by about 0.028 u, so each fusion releases about 26 MeV. The Sun converts roughly 600 million tonnes of hydrogen per second, and "burns" about 4 million tonnes of mass into pure energy every second, in perfect agreement with $E = mc^2$. The light that warms the farms of the North West Region left the Sun as mass defect!

EXERCISE 3 **Uranium-235 and the Aston curve**

Given $m(^{235}_{92}\mathrm{U}) = 234.9934$ u (nucleus), $m_p = 1.00728$ u, $m_n = 1.00866$ u.

1. Calculate the mass defect and the binding energy of the uranium-235 nucleus in MeV.
2. Deduce its binding energy per nucleon, and compare with iron-56 (8.8 MeV/nucleon). What do you conclude about the possibility of releasing energy by fission of uranium-235?

SOLUTION **Exercise 1**

1. Uranium-235 has $Z = 92$ protons and $A - Z = 143$ neutrons:

$$\Delta m = 92(1.00728) + 143(1.00866) - 234.9934 = 92.6698 + 144.2384 - 234.9934 = 1.9148 \text{ u}.$$

$$E_l = 1.9148 \times 931.5 \approx 1783.6 \text{ MeV} \approx 1.78 \text{ GeV}.$$

2. Binding energy per nucleon:

$$\frac{E_l}{A} = \frac{1783.6}{235} \approx 7.6 \text{ MeV/nucleon}.$$

This is well below the 8.8 MeV/nucleon of iron-56: uranium-235 lies on the descending branch of the Aston curve. If it splits into two medium-mass fragments (each with $E_l/A \approx 8.5$ MeV/nucleon), each nucleon gains about 0.9 MeV of binding, so the fission of one nucleus releases roughly $235 \times 0.9 \approx 200$ MeV: fission is exoenergetic, as we shall study in the next section.

KEY POINTS **Key points of this section**

- Masses: $m_p = 1.00728$ u, $m_n = 1.00866$ u; $1\text{ u} = 1.66054 \times 10^{-27}$ kg.
- Mass defect: $\Delta m = [Zm_p + (A - Z)m_n] - m_{\text{nucleus}} > 0$.

- Einstein: $E = mc^2$; binding energy $E_l = \Delta m\, c^2$; conversion 1 u ↔ 931.5 MeV.
- Stability is measured by E_l/A; maximum ≈ 8.8 MeV/nucleon at iron-56 (Aston curve).
- Energy balance: $Q = (m_{\text{reactants}} - m_{\text{products}})c^2$; $Q > 0$ exoenergetic, $Q < 0$ endoenergetic.

We now understand why certain nuclei can release energy: heavy nuclei by splitting, light nuclei by merging. The next section examines these two processes in detail: nuclear fission, harnessed in power reactors, and nuclear fusion, the energy source of the stars.

4 NUCLEAR FISSION AND NUCLEAR FUSION

In the previous section we established Einstein's mass-energy relation $E = \Delta m\, c^2$ and the **Aston curve** of binding energy per nucleon. We saw that the curve rises steeply for light nuclei, reaches a maximum near iron ($A \approx 56$, about 8.8 MeV per nucleon), and then decreases gently for heavy nuclei. This shape has a spectacular practical consequence: there are two ways of rearranging nucleons so that the products sit higher on the curve (more tightly bound), and therefore release energy. A very heavy nucleus can split into medium-sized fragments — this is **fission**. Two very light nuclei can join into a heavier one — this is **fusion**. These two processes power nuclear reactors, the Sun, and the most destructive weapons ever built. This section studies them in turn and then compares them.

4.1 Nuclear fission

Observation. In 1938, Otto Hahn and Fritz Strassmann bombarded uranium with neutrons and were astonished to find barium — a much lighter element — among the products. Lise Meitner and Otto Frisch correctly interpreted the result: the uranium nucleus had been split in two. How can a slow, electrically neutral particle break apart a nucleus containing 92 protons? Precisely because it is neutral, the neutron is not repelled by the nuclear charge, so it can approach and be absorbed; the resulting compound nucleus ${}^{236}_{92}\mathrm{U}$ is highly unstable and deforms like a liquid drop until it breaks into two pieces.

DEFINITION **Nuclear fission**

Nuclear fission is a nuclear reaction in which a heavy nucleus ($A \gtrsim 230$) splits into two lighter nuclei (the fission fragments), with the emission of a few neutrons and the release of a large amount of energy. Fission of ${}^{235}_{92}\mathrm{U}$ is usually induced by the capture of a slow (thermal) neutron, of kinetic energy of order 0.025 eV. (A few very heavy nuclei also undergo spontaneous fission, without any projectile, but this is rare.)

A typical induced fission of uranium-235 is:

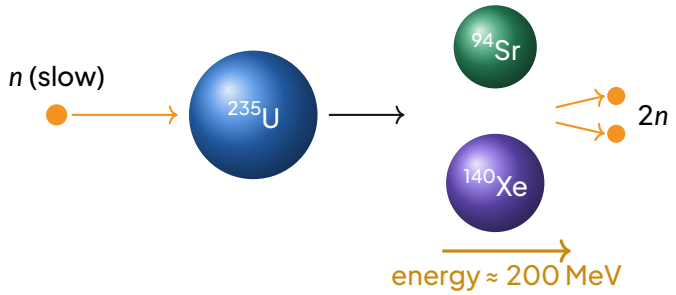
$$ {}^{235}_{92}\mathrm{U} + {}^{1}_{0}\mathrm{n} \longrightarrow {}^{94}_{38}\mathrm{Sr} + {}^{140}_{54}\mathrm{Xe} + 2\,{}^{1}_{0}\mathrm{n} $$

Another frequently observed channel is:

$$^{235}_{92}\mathrm{U} + {}^{1}_{0}\mathrm{n} \longrightarrow {}^{92}_{36}\mathrm{Kr} + {}^{141}_{56}\mathrm{Ba} + 3\,{}^{1}_{0}\mathrm{n}$$

WARNING **Fission is not unique**

Uranium-235 can split in many different ways (more than thirty fragment pairs are known). All fission equations must conserve charge number Z and nucleon number A. Check the first equation: 235 + 1 = 94 + 140 + 2 = 236 ✓ and 92 + 0 = 38 + 54 + 0 = 92 ✓. The number of emitted neutrons depends on the channel (usually 2 or 3). Note also that the fragments are generally radioactive: being neutron-rich, they decay by β^- emission until they reach stability — this is why used reactor fuel remains dangerous for a very long time.



Induced fission of uranium-235 by a slow neutron: two fragments, two or three neutrons and about 200 MeV of energy are released.

Why does fission release energy? The fragments (Sr, Xe, Kr, Ba, …) have A in the range 90–145, a region where the binding energy per nucleon (≈ 8.5 MeV) is greater than that of uranium (≈ 7.6 MeV). The products are therefore more tightly bound: the total mass of the products is less than the initial mass, and the difference appears as kinetic energy of the fragments and neutrons, plus γ radiation. A detailed mass balance gives about 200 MeV per fission — roughly 3.2×10^{-11} J per nucleus, which is enormous compared with the few electron-volts released per atom in a chemical reaction.

4.2 The chain reaction and critical mass

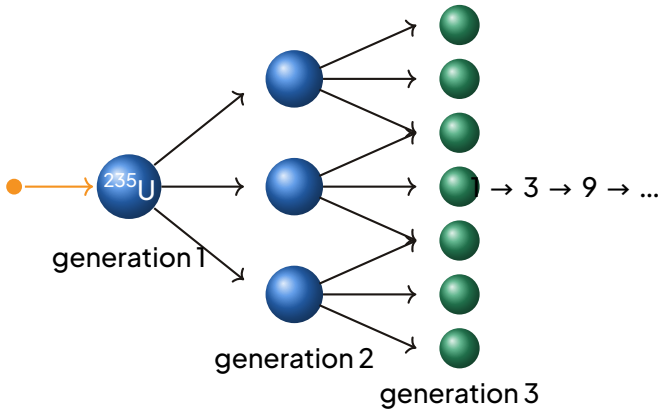
Observation. Each fission needs one neutron but produces two or three. If these neutrons in turn strike other ^{235}U nuclei, the number of fissions multiplies generation after generation: 1 → 3 → 9 → 27 → … This self-sustaining process is the **chain reaction**.

DEFINITION **Multiplication factor and critical mass**

The **multiplication factor** k is the ratio of the number of fissions in one generation to the number in the previous generation. The **critical mass** is the minimum mass of fissile material needed for a self-sustaining chain reaction (k = 1): below it, too many neutrons escape

through the surface or are captured uselessly, and the reaction dies out.

Regime	Condition	Consequence
Subcritical	$k < 1$	Reaction dies out.
Critical	$k = 1$	Steady, controlled power (nuclear reactor).
Supercritical	$k > 1$	Explosive growth (atomic bomb).



Chain reaction: neutrons from each fission trigger further fissions, so the number of reactions grows geometrically.

In an **atomic bomb**, two subcritical masses of ^{235}U (or plutonium-239) are slammed together to form a highly supercritical mass: the chain reaction runs away in about a microsecond, releasing an enormous amount of energy. In a **nuclear reactor**, the goal is exactly the opposite: keep $k = 1$ permanently so that energy is produced at a constant, controlled rate.

Main parts of a pressurised water reactor (PWR).

- **Fuel:** rods of uranium slightly enriched in ^{235}U, where fission takes place.
- **Moderator** (ordinary water): slows the fast fission neutrons down to thermal speeds, because slow neutrons are far more likely to induce further fission of ^{235}U.
- **Control rods** (cadmium or boron): absorb neutrons. Pushing them in lowers k; withdrawing them raises k. They keep the reactor exactly critical and allow emergency shutdown.
- **Coolant:** the pressurised water also carries the heat away to a steam generator; the steam drives a turbine coupled to an alternator, exactly as in a thermal power station.
- **Containment:** a thick steel and concrete vessel that protects the environment from radiation and radioactive leaks.

DID YOU KNOW? **Nuclear energy and Africa**

Only South Africa currently operates a commercial nuclear power station (Koeberg, near Cape Town), but several African countries, including Ghana, Nigeria and Egypt, are developing nuclear programmes, and many operate small research reactors used for medicine, agriculture and materials analysis. Cameroon, whose electricity demand grows every year and whose grid relies heavily on hydroelectric dams vulnerable to drought, takes part in international training and follows International Atomic Energy Agency (IAEA) standards for radiation safety and radioactive waste management. Any future nuclear option would require strict solutions for the long-term storage of highly radioactive fission products.

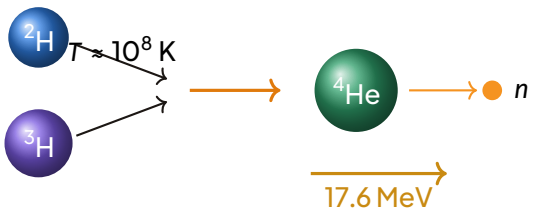
4.3 Nuclear fusion

Observation. The Sun has been radiating about 4×10^{26} W for billions of years. No chemical or fission fuel could last so long. Its energy comes from the opposite process: light nuclei (hydrogen) fusing into heavier ones (helium).

DEFINITION **Nuclear fusion**

Nuclear fusion is a nuclear reaction in which two light nuclei unite to form a heavier nucleus, with the emission of a particle (often a neutron) and the release of energy. Example (the deuterium–tritium reaction):

$$^{2}_{1}\mathrm{H} + ^{3}_{1}\mathrm{H} \longrightarrow ^{4}_{2}\mathrm{He} + ^{1}_{0}\mathrm{n}$$



Fusion of deuterium and tritium into helium-4, releasing a neutron and 17.6 MeV.

Why is fusion so difficult to achieve? Both nuclei are positively charged, so they repel each other electrically (Coulomb repulsion). To bring them close enough ($\approx 10^{-15}$ m) for the strong nuclear force to bind them, they must collide with enormous kinetic energy. This requires temperatures of about 10^{8} K, at which matter is a plasma (fully ionised gas) that no ordinary container can hold. In **stars**, gravity provides the confinement and the core temperature ignites the fusion of hydrogen into helium. In the **hydrogen bomb**, a small fission bomb is used as a match to reach the ignition temperature — an uncontrolled fusion. Controlled fusion on Earth is the goal of the international **ITER** project (France), which confines the plasma with powerful magnetic fields; it aims to demonstrate that fusion can produce more energy than it consumes, but commercial fusion power remains a long-term hope.

PROPERTY **Fission and fusion both release energy**

Both the fission of a heavy nucleus and the fusion of light nuclei release energy because, in each case, the products lie higher on the Aston curve (greater binding energy per nucleon) than the reactants. The total mass decreases and the mass defect appears as energy, $E = \Delta m\, c^2$. (Not a proof to reproduce in the examination, but you must be able to justify it from the curve.)

METHOD **Energy released by a fission or fusion reaction**

1. Write the balanced nuclear equation (conserve A and Z).
2. Compute the mass defect: $\Delta m = m_{\text{reactants}} - m_{\text{products}}$ (use the atomic or nuclear masses as given in the question).
3. Convert to energy: $E = \Delta m\, c^2$, or directly E (MeV) = Δm (u) × 931.5 MeV/u.
4. Quote the result with its unit; a positive Δm means energy is released.

WORKED EXAMPLE 6 **Energy released by the D–T fusion reaction**

Given $m(^2\text{H}) = 2.0141$ u, $m(^3\text{H}) = 3.0160$ u, $m(^4\text{He}) = 4.0026$ u, $m_n = 1.0087$ u and 1 u ≡ 931.5 MeV, calculate the energy released in $^2\text{H} + {}^3\text{H} \longrightarrow {}^4\text{He} + \text{n}$.

SOLUTION

Solution. Mass of reactants: 2.0141 + 3.0160 = 5.0301 u. Mass of products: 4.0026 + 1.0087 = 5.0113 u. Mass defect:

$$\Delta m = 5.0301 - 5.0113 = 0.0188 \text{ u}.$$

Energy released:

$$E = 0.0188 \times 931.5 \approx 17.5 \text{ MeV} \approx 2.8 \times 10^{-12} \text{ J}.$$

This is smaller than the ≈ 200 MeV of one fission, but per kilogram of fuel fusion releases several times more energy, because the reacting nuclei are so light.

WORKED EXAMPLE 7 **Balancing a fission equation**

Complete the fission equation $^{235}_{92}\text{U} + {}^{1}_{0}\text{n} \rightarrow {}^{92}_{36}\text{Kr} + {}^{141}_{56}\text{Ba} + x\,{}^{1}_{0}\text{n}$ by finding x.

SOLUTION

Solution. Conservation of nucleon number:

$$235 + 1 = 92 + 141 + x \quad \Rightarrow \quad x = 236 - 233 = 3.$$

Conservation of charge: $92 = 36 + 56$ ✓. Hence three neutrons are emitted.

4.4 Fission versus fusion: a comparison

Criterion	Fission	Fusion
Fuel	^{235}U, ^{239}Pu: scarce, mined	^{2}H from sea water: almost unlimited
Energy per kg of fuel	Very high	Even higher (several times)
Waste	Long-lived radioactive fission products	Little radioactive waste (activated structures)
Safety	Risk of runaway ($k > 1$), meltdown	No chain reaction; reaction stops if confinement fails
Conditions	Achieved at ordinary temperatures	Needs $T \approx 10^8$ K: not yet industrial
Uses	Power reactors, atomic bomb	Stars, hydrogen bomb; ITER research

WARNING **Common mistakes**

- Saying that fusion occurs in today's nuclear power plants. All operating reactors use fission; controlled fusion is still experimental (ITER).
- Confusing the critical mass with the total mass of fuel: the critical mass is the minimum mass for a self-sustaining chain reaction, not the amount loaded into a reactor.
- Forgetting that fission equations have several possible fragment pairs — never "correct" the given fragments; only find the number of neutrons.
- Mixing up the two reactions: fission = heavy nucleus splits; fusion = light nuclei join.
- Writing $E = \Delta m\, c^2$ with Δm in atomic mass units and c in SI units without converting: either convert Δm to kilograms, or use the shortcut $1\ \text{u} \equiv 931.5$ MeV.

METHOD **Exam tip: finding the number of emitted neutrons**

In a fission equation with unknown neutron number x, apply conservation of nucleon number: $x = (A_{\text{fuel}} + 1) - (A_{\text{fragment 1}} + A_{\text{fragment 2}})$, then verify that charge is conserved. This is a frequent two-mark question at the GCE.

KEY POINTS **Key points**

- Fission: a heavy nucleus (^{235}U) captures a slow neutron and splits into two fragments plus 2–3 neutrons, releasing ≈ 200 MeV.
- The emitted neutrons sustain a chain reaction; $k = 1$ in a reactor (controlled), $k > 1$ in a bomb (uncontrolled); a critical mass is required.
- A PWR uses fuel rods, a moderator, control rods, a coolant and a containment vessel.
- Fusion: two light nuclei (^{2}H + ^{3}H) unite at $T \approx 10^{8}$ K; it powers the stars and the hydrogen bomb; ITER seeks controlled fusion.
- Both processes release energy because the products are higher on the Aston curve; compute $E = \Delta m\, c^{2}$ with Δm in u and 1 u ≡ 931.5 MeV.

EXERCISE 4 **Energy from uranium fission**

One fission of ^{235}U releases on average 200 MeV.

1. Express this energy in joules.
2. Calculate the energy released by the complete fission of 1.0 g of ^{235}U (molar mass 235 g mol^{-1}, $N_A = 6.02 \times 10^{23}$ mol^{-1}).
3. Compare with the combustion of petrol (≈ 4.5×10^{7} J kg^{-1}) and comment.

SOLUTION **Exercise 1**

1. $E = 200 \times 10^{6} \times 1.60 \times 10^{-19} = 3.2 \times 10^{-11}$ J.
2. Number of nuclei in 1.0 g:

$$N = \frac{1.0}{235} \times 6.02 \times 10^{23} = 2.56 \times 10^{21} \text{ nuclei.}$$

Total energy: $E_{\text{tot}} = 2.56 \times 10^{21} \times 3.2 \times 10^{-11} \approx 8.2 \times 10^{10}$ J.

3. Burning 1 g of petrol releases about 4.5×10^{4} J. Fission of the same mass of uranium releases roughly two million times more energy, which is why nuclear fuel is so concentrated — and why its handling demands extreme care.

EXERCISE 5 **Energy released in a fission reaction**

Consider the fission $^{235}_{92}\mathrm{U} + {}^{1}_{0}\mathrm{n} \longrightarrow {}^{94}_{38}\mathrm{Sr} + {}^{140}_{54}\mathrm{Xe} + 2\,{}^{1}_{0}\mathrm{n}$. Given $m(^{235}\mathrm{U}) = 235.0439\ \mathrm{u}$, $m(^{94}\mathrm{Sr}) = 93.9154\ \mathrm{u}$, $m(^{140}\mathrm{Xe}) = 139.9252\ \mathrm{u}$, $m_n = 1.0087\ \mathrm{u}$ and $1\ \mathrm{u} \equiv 931.5\ \mathrm{MeV}$:

1. Calculate the mass defect Δm for this reaction.
2. Deduce the energy released per fission, in MeV, and compare with the average value of 200 MeV.

SOLUTION **Exercise 2**

1. Mass of reactants: $235.0439 + 1.0087 = 236.0526\ \mathrm{u}$. Mass of products: $93.9154 + 139.9252 + 2 \times 1.0087 = 235.8580\ \mathrm{u}$.

$$\Delta m = 236.0526 - 235.8580 = 0.1946\ \mathrm{u}.$$

2. $E = 0.1946 \times 931.5 \approx 181\ \mathrm{MeV}$. This is of the same order as the average 200 MeV; the exact value depends on the fission channel, and part of the energy is carried away later by the radioactive decay of the fragments.

5 INTEGRATION ACTIVITY

5.1 Aims of the activity

This integration activity places you in the role of energy consultants for **ENEO** (Energy of Cameroon), the national electricity utility. You will mobilise the full toolkit of nuclear physics developed in this chapter — nuclear equations, mass defect calculations, binding energy curves, and radiation safety principles — to evaluate two nuclear options for diversifying Cameroon's energy mix, currently dominated by hydroelectricity. By the end of this activity, you should be able to:

- Balance nuclear reactions for fission and fusion using conservation of nucleon number A and charge (proton) number Z.
- Calculate the energy released per reaction and per unit mass of fuel using the mass–energy equivalence $\Delta E = \Delta m c^2$ and the conversion $1\,\text{u} = 931.5\,\text{MeV}/c^2$.
- Interpret the Aston curve (binding energy per nucleon versus mass number) to explain qualitatively why both very heavy and very light nuclei can release energy.
- Synthesise technical, safety, environmental and economic arguments into a concise policy memo suitable for decision-makers.

5.2 Situation

Cameroon's electricity demand is growing steadily, by several per cent per year. While hydroelectric projects (Lom Pangar, Nachtigal, Memve'ele) provide a low-carbon base, seasonal variability of river flows and the need for firm baseload power lead planners to ask whether nuclear energy could one day play a role. You are part of a multidisciplinary team of young physicists and engineers seconded to **ENEO**'s Strategic Planning Division. Your mandate: produce a one-page technical memo comparing the viability of a **Pressurised Water Reactor (PWR) fission plant** fuelled by enriched uranium versus a hypothetical **Deuterium–Tritium (D–T) fusion pilot plant** for the Yaoundé region (peak demand ~ 400 MW).

The following nuclear data are provided (atomic masses in unified atomic mass units, u):

Particle / Nucleus	Atomic mass (u)
Neutron (^1_0n)	1.008 665
Uranium-235 ($^{235}_{92}$U)	235.043 930
Krypton-95 ($^{95}_{36}$Kr)	94.939 710
Barium-140 ($^{140}_{56}$Ba)	139.910 581
Deuterium ($^2_1$H)	2.014 102
Tritium ($^3_1$H)	3.016 049
Helium-4 ($^4_2$He)	4.002 603

Constants: $1\,\text{u} = 931.5\,\text{MeV}/c^2$; $N_A = 6.022 \times 10^{23}\,\text{mol}^{-1}$; $1\,\text{eV} = 1.602 \times 10^{-19}\,\text{J}$, so $1\,\text{MeV} = 1.602 \times 10^{-13}\,\text{J}$.

5.3 Tasks

1. **Balance the nuclear equations.**
   (a) Write the balanced equation for the induced fission of $^{235}_{92}\mathrm{U}$ by a thermal neutron producing $^{95}_{36}\mathrm{Kr}$, $^{140}_{56}\mathrm{Ba}$ and x neutrons. Determine x.
   (b) Write the balanced equation for the fusion of deuterium and tritium ($^{2}_{1}\mathrm{H} + ^{3}_{1}\mathrm{H} \rightarrow ^{4}_{2}\mathrm{He} + n$). Verify the conservation laws.
2. **Energy yield calculations.**
   (a) Calculate the mass defect Δm (in u) and the energy Q (in MeV) released per reaction for both the fission and the fusion processes above.
   (b) Calculate the energy released per kilogram of fuel for each process. For fission, assume pure $^{235}\mathrm{U}$ fuel. For fusion, assume an equimolar D–T mixture (effective molar mass $\approx 5.03\ \mathrm{g\,mol^{-1}}$).
   (c) Compare the specific energy densities (in $\mathrm{J\,kg^{-1}}$) and comment on the orders of magnitude.
3. **Aston curve justification.** Using the general shape of the binding energy per nucleon (B/A) curve (Aston curve), explain why both reactions are exothermic. Sketch a qualitative B/A versus A graph indicating the approximate positions of the initial and final nuclei for each reaction. Show that the products lie higher on the curve (more tightly bound) than the reactants.
4. **Policy memo (one page max).** Draft a memo to the Director General of ENEO recommending an energy strategy. Your memo must include:
   - A clear recommendation (Fission now / Fusion R&D / Neither / Hybrid).
   - Quantitative support from Tasks 2 and 3.
   - Radiation safety considerations: nature of waste (half-lives, activity), shielding requirements, and proliferation risk.
   - Cameroonian context: uranium availability in the Central African region, cooling water constraints (Sanaga river), grid stability, regulatory capacity, and public acceptance.

   Groups will present and defend their memo in a 10–minute plenary session.

5.4 Model answer

Task 1: Balancing the nuclear equations

METHOD **Balancing any nuclear equation**

1. Write the reaction with unknown coefficients.
2. Conserve the **nucleon number** A (top numbers): sum of A on the left = sum of A on the right.
3. Conserve the **charge number** Z (bottom numbers): sum of Z on the left = sum of Z on

the right.

4. Solve for the unknown(s) and rewrite the final balanced equation.

1. **Fission:** Conservation of nucleon number A and proton number Z:

$$^{235}_{92}\mathrm{U} + ^{1}_{0}n \rightarrow ^{95}_{36}\mathrm{Kr} + ^{140}_{56}\mathrm{Ba} + x\,^{1}_{0}n$$

$$A: \quad 235 + 1 = 95 + 140 + x \implies 236 = 235 + x \implies \boxed{x = 1}$$
$$Z: \quad 92 + 0 = 36 + 56 + 0 \implies 92 = 92 \quad \text{(conserved)}$$

Balanced equation:

$$^{235}_{92}\mathrm{U} + ^{1}_{0}\mathrm{n} \longrightarrow ^{95}_{36}\mathrm{Kr} + ^{140}_{56}\mathrm{Ba} + ^{1}_{0}\mathrm{n}$$

WARNING **Common mistake**

Many candidates assume $x = 2$ or 3 because on average a ^{235}U fission event releases 2–3 neutrons. Here the specific fragment pair (^{95}Kr, ^{140}Ba) fixes $x = 1$ by nucleon conservation. Always balance the equation as given, not the statistical average.

2. **Fusion:**

$$^{2}_{1}\mathrm{H} + ^{3}_{1}\mathrm{H} \rightarrow ^{4}_{2}\mathrm{He} + ^{1}_{0}n$$

Check: A: 2 + 3 = 4 + 1 = 5; Z: 1 + 1 = 2 + 0 = 2. Both conserved.

Task 2: Energy yield calculations

KEY POINTS **Recipe for a Q-value**

- $\Delta m = m_{\text{initial}} - m_{\text{final}}$ (mass that "disappears").
- $Q = \Delta m \times 931.5\ \text{MeV/u}$ if masses are in u, or $Q = \Delta m\, c^2$ in SI.
- If $\Delta m > 0$, the reaction is **exothermic** (releases energy).

1. **Mass defect and Q-value per reaction.**

Fission:

$$m_{\text{initial}} = m(^{235}\mathrm{U}) + m_n = 235.043\,930 + 1.008\,665 = 236.052\,595\ \text{u}$$
$$m_{\text{final}} = m(^{95}\mathrm{Kr}) + m(^{140}\mathrm{Ba}) + m_n$$
$$= 94.939\,710 + 139.910\,581 + 1.008\,665 = 235.858\,956\ \text{u}$$
$$\Delta m_{\text{fis}} = m_{\text{initial}} - m_{\text{final}} = 236.052\,595 - 235.858\,956 = 0.193\,639\ \text{u}$$
$$Q_{\text{fis}} = \Delta m_{\text{fis}} \times 931.5\ \text{MeV/u} = 0.193\,639 \times 931.5 \approx \boxed{180.4\ \text{MeV}}$$

Fusion:

$$m_{\text{initial}} = m(^2\text{H}) + m(^3\text{H}) = 2.014\,102 + 3.016\,049 = 5.030\,151\ \text{u}$$
$$m_{\text{final}} = m(^4\text{He}) + m_n = 4.002\,603 + 1.008\,665 = 5.011\,268\ \text{u}$$
$$\Delta m_{\text{fus}} = 5.030\,151 - 5.011\,268 = 0.018\,883\ \text{u}$$
$$Q_{\text{fus}} = 0.018\,883 \times 931.5 \approx \boxed{17.59\ \text{MeV}}$$

Comment. One fission releases about ten times more energy than one fusion per reaction — but a uranium nucleus is about 47 times heavier than a D–T pair. Per unit mass of fuel, fusion will win: this is what we check next.

2. **Energy per kilogram of fuel.**

Fission (pure ^{235}U): Molar mass $M_U \approx 235\ \text{g mol}^{-1} = 0.235\ \text{kg mol}^{-1}$. Number of nuclei per kg: $N = \dfrac{N_A}{M_U} = \dfrac{6.022 \times 10^{23}}{0.235} = 2.563 \times 10^{24}\ \text{kg}^{-1}$.

$$E_{\text{fis, kg}} = N \times Q_{\text{fis}} \times 1.602 \times 10^{-13}\ \text{J/MeV}$$
$$= 2.563 \times 10^{24} \times 180.4 \times 1.602 \times 10^{-13}$$
$$\approx \boxed{7.41 \times 10^{13}\ \text{J kg}^{-1}}$$

Fusion (D–T mixture, $M_{\text{mix}} \approx 5.03\ \text{g mol}^{-1}$): One reaction consumes one D and one T nucleus, i.e. one "mole of reactions" consumes 5.03 g of equimolar mixture. Number of reactions per kg: $N' = \dfrac{N_A}{0.00503} = \dfrac{6.022 \times 10^{23}}{5.03 \times 10^{-3}} = 1.197 \times 10^{26}\ \text{kg}^{-1}$.

$$E_{\text{fus, kg}} = N' \times Q_{\text{fus}} \times 1.602 \times 10^{-13}$$
$$= 1.197 \times 10^{26} \times 17.59 \times 1.602 \times 10^{-13}$$
$$\approx \boxed{3.37 \times 10^{14}\ \text{J kg}^{-1}}$$

3. **Comparison:**

$$\frac{E_{\text{fus, kg}}}{E_{\text{fis, kg}}} \approx \frac{3.37 \times 10^{14}}{7.41 \times 10^{13}} \approx 4.6$$

Fusion releases about **4.6 times more energy per unit mass of fuel** than fission. Both are roughly 10^6–10^7 times more energy-dense than chemical combustion (petrol $\approx 4.6 \times 10^7\ \text{J kg}^{-1}$; coal $\approx 3 \times 10^7\ \text{J kg}^{-1}$).

Task 3: Aston curve justification

KEY POINTS **The Aston Curve (Binding Energy per Nucleon)**

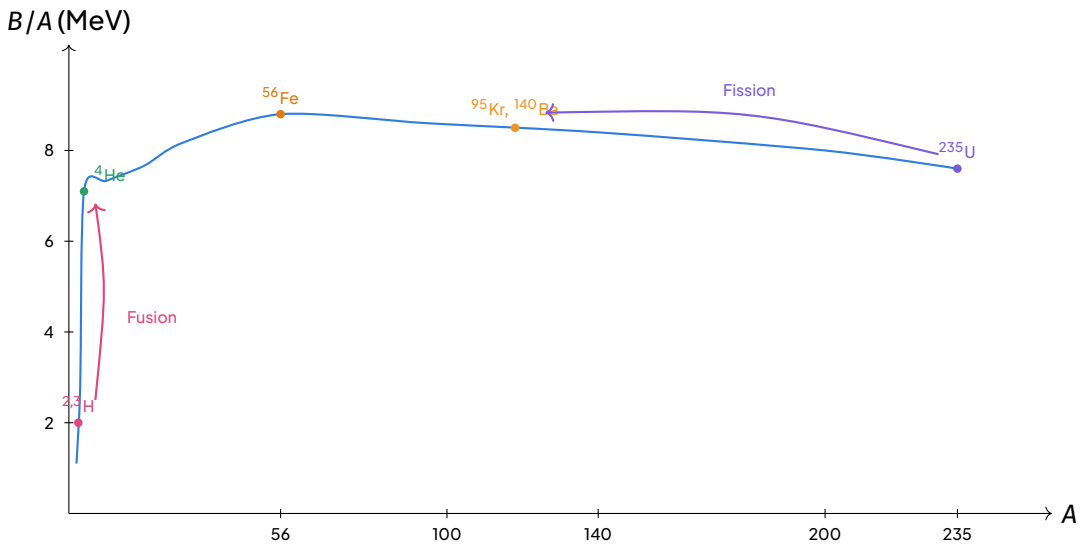
The curve of B/A versus A peaks near ^{56}Fe ($B/A \approx 8.8$ MeV per nucleon), the most tightly bound region of the nuclear chart.

- Fission: a heavy nucleus ($A \approx 235$, $B/A \approx 7.6$ MeV) splits into medium-mass fragments

($A \approx 95$ and 140, $B/A \approx 8.5$ MeV). The products are more tightly bound ⇒ energy is released.

- Fusion: very light nuclei ($A = 2, 3$; $B/A \approx 1.1$ and 2.8 MeV) fuse into ^{4}He ($B/A \approx 7.1$ MeV). The product is much more tightly bound ⇒ large energy release.

In both cases the reaction moves the system towards the peak at $A \approx 56$, increasing the average binding energy per nucleon. The mass defect Δm corresponds exactly to this gain in total binding energy: $\Delta mc^2 = B_{\text{products}} - B_{\text{reactants}}$.



Qualitative Aston curve (horizontal scale: $x = A/20$). Fusion (light → medium) and fission (heavy → medium) both climb towards the iron peak, i.e. towards more tightly bound nuclei, releasing energy.

Task 4: Policy Memo — Model Response

MEMORANDUM
TO: Director General, ENEO
FROM: Nuclear Physics Advisory Group, Upper Sixth Cohort
SUBJECT: Nuclear Energy Option for Yaoundé Baseload — Fission vs. Fusion

1. Executive Recommendation We recommend a **phased strategy**: a **feasibility study for a Small Modular Reactor (SMR) fission plant** as a long-term option, while devoting a small fraction of the energy R&D budget to **international fusion partnerships**. A domestic fusion pilot plant is not realistic in the foreseeable future: no fusion reactor anywhere in the world yet produces net electricity.

2. Quantitative Energy Density Argument Our calculations (Task 2) give:

- Fission (^{235}U): 7.4×10^{13} J kg^{-1}.
- Fusion (D–T): 3.4×10^{14} J kg^{-1}, i.e. ≈ 4.6 times more per kg.

For a 400 MW$_e$ plant at ~ 33% thermal efficiency, the thermal power is ≈ 1.2×10^9 W, i.e. ≈ 1.0×10^{14} J per day. Fuel consumption (assuming complete burn-up):

- Fission: $\frac{1.0 \times 10^{14}}{7.4 \times 10^{13}} \approx 1.4$ kg of ^{235}U per day.
- Fusion: $\frac{1.0 \times 10^{14}}{3.4 \times 10^{14}} \approx 0.3$ kg of D–T mixture per day.

Either way, a few hundred kilograms of fuel per year replace millions of tonnes of fossil fuel. The decisive difference is **technological maturity**: PWRs are proven, licensed technology with hundreds of units operating worldwide; D–T fusion remains at the experimental stage (research tokamaks; first net-energy demonstrations only very recent and not yet at plant scale).

3. Radiation Safety & Waste Profile

Aspect	Fission (PWR)	Fusion (D–T)
High-level waste	Spent fuel contains long-lived actinides (e.g. ^{239}Pu, $T_{1/2} \approx 24\,000$ yr) and fission products (^{137}Cs, ^{90}Sr, $T_{1/2} \approx 30$ yr). Requires deep geological storage.	No actinides. Structural materials activated by 14.1 MeV neutrons; with low-activation alloys, waste decays to safe levels in about a century.
Radioactive inventory	Large (core + storage pool). Residual decay heat requires active cooling long after shutdown.	Moderate (tritium inventory of order kilograms, activated structures). No runaway decay-heat scenario.
Proliferation	Enrichment and reprocessing are dual-use technologies; international safeguards mandatory.	No fissile material; tritium handling is the main concern. Lower proliferation profile.
Shielding	Thick concrete and steel against γ rays and neutrons.	Thick blanket/shield against 14.1 MeV neutrons; comparable footprint.

4. Cameroonian Context

- **Fuel security:** uranium deposits are known in the Central African region (e.g. Bakouma in neighbouring CAR); a regional supply partnership would be needed since Cameroon has no enrichment industry. Deuterium is extractable from water; tritium must be bred from lithium — a technology not yet demonstrated at scale.
- **Cooling:** a ~ 1.2 GW$_{th}$ plant needs a few tens of m^3 s^{-1} of cooling water; the Sanaga

WARNING **Examiner's traps in the plenary defence**

- Atomic vs. nuclear masses: the given masses are atomic masses (they include the electrons). In the fission and fusion equations above, the electron numbers balance automatically (92 e^- on each side for fission; 2 e^- on each side for fusion), so atomic masses can be used directly. In β^+ decay or electron capture this is not automatic — always check electron balance.
- Per kg of what? Fission: per kg of ^{235}U. Fusion: per kg of the D–T mixture. Do not quote per kg of tritium alone.
- Aston curve axes: the x-axis is the mass number A, not the atomic number Z; the peak is at $A \approx 56$ (Fe/Ni region).
- Units: energy per reaction in MeV; energy per kg in J kg^{-1}. Convert with 1 MeV = 1.602×10^{-13} J.
- Sign of Δm: define $\Delta m = m_{initial} - m_{final}$ so that an exothermic reaction gives $\Delta m > 0$ and $Q > 0$. Mixing up the order flips the sign and costs marks.

6 METHODS AND WORKED EXAMPLES

7 ATOMIC NUCLEUS AND NUCLEAR STABILITY

DID YOU KNOW? **Discovery of the Nucleus**

In 1911, Ernest Rutherford, Hans Geiger and Ernest Marsden performed the famous gold foil experiment at Manchester. They observed that while most α particles passed through a thin gold foil with little deflection, a few were scattered at large angles ($> 90^\circ$). Rutherford concluded: "It was quite the most incredible event that has ever happened to me in my life. It was almost as incredible as if you fired a 15-inch shell at a piece of tissue paper and it came back and hit you." This led to the nuclear model: a tiny, dense, positively charged nucleus containing almost all the mass, surrounded by electrons.

DEFINITION **Nuclear Composition and Notation**

An atomic nucleus is characterised by:

- Proton number (Atomic number) Z: Number of protons. Determines the chemical element.
- Neutron number N: Number of neutrons.
- Nucleon number (Mass number) $A = Z + N$: Total number of nucleons (protons + neutrons).

A nuclide is denoted ${}^{A}_{Z}\mathrm{X}$ (e.g., ${}^{238}_{92}\mathrm{U}$). Isotopes have same Z, different N (and A). Isotones have same N. Isobars have same A.

DEFINITION **Unified Atomic Mass Unit**

The unified atomic mass unit (u) is defined as $\frac{1}{12}$ the mass of a neutral ${}^{12}_{6}\mathrm{C}$ atom.

$$1\,\mathrm{u} = 1.660539e-27\,\mathrm{kg} = 931.494\,\mathrm{MeV}/c^2$$

Masses of fundamental particles: $m_p = 1.007276\,\mathrm{u}$, $m_n = 1.008665\,\mathrm{u}$, $m_e = 0.00054858\,\mathrm{u}$. Atomic mass M_{atom} includes Z electrons: $M_{\mathrm{atom}} \approx M_{\mathrm{nucleus}} + Zm_e - B_e/c^2$ (electron binding energy B_e negligible).

PROPERTY **Nuclear Stability and the N–Z Curve**

- Strong Nuclear Force: Short-range ($\approx 1-3$ fm), attractive, charge-independent (acts $p-p, n-n, p-n$), saturates. Overcomes Coulomb repulsion between protons.
- Stability Belt: For light nuclei ($A < 40$), stable nuclei have $N \approx Z$. For heavier nuclei, Coulomb repulsion requires excess neutrons ($N > Z$) to provide more strong force without adding repulsion. The line of stability curves upward.
- Even-Even preference: Nuclei with even Z and even N are most stable (pairing effect). Odd-odd nuclei are rare (only 4 stable: ${}^{2}_{1}\mathrm{H}, {}^{6}_{3}\mathrm{Li}, {}^{10}_{5}\mathrm{B}, {}^{14}_{7}\mathrm{N}$).
- Magic Numbers: Z or $N = 2, 8, 20, 28, 50, 82, 126$ confer extra stability (shell model).

EXPERIMENT **The Segre Chart (N vs Z)**

Plot N vs Z for known nuclides. Stable nuclides (black dots) form a narrow "valley of stability". Neutron-rich nuclei lie above → β^- decay. Proton-rich nuclei lie below → β^+ decay or Electron Capture (EC). Very heavy nuclei ($Z > 82$) → α decay or spontaneous fission. Technetium ($Z = 43$) and Promethium ($Z = 61$) have no stable isotopes.

PROPERTY **Binding Energy and Mass Defect**

The mass of a nucleus is less than the sum of its free constituent nucleons. The difference is the mass defect Δm.

$$\Delta m = \left[Z m_p + (A-Z) m_n\right] - M_{\text{nucleus}}$$

Using atomic masses (includes electrons, cancels out if Z conserved):

$$\Delta m = \left[Z M({}^{1}_{1}\mathrm{H}) + (A-Z) m_n\right] - M({}^{A}_{Z}\mathrm{X})$$

The Binding Energy E_b is the energy equivalent (Einstein):

$$E_b = \Delta m\, c^2$$

Binding Energy per Nucleon E_b/A measures stability. Maximum $\approx$ 8.8 MeV at ${}^{56}_{26}\mathrm{Fe}$ (most stable nucleus).

KEY POINTS **Binding Energy Curve Features**

- Rises steeply for light nuclei → Fusion (combining light nuclei) releases energy (prod-

ucts more tightly bound).

- Broad maximum around $A = 56$ (Fe, Ni).
- Falls slowly for heavy nuclei → Fission (splitting heavy nuclei) releases energy (fragments more tightly bound).
- $^4_2$He (α particle) has exceptionally high $E_b/A \approx 7.07$ MeV → explains α decay prevalence in heavy nuclei.

WORKED EXAMPLE 8 **Binding Energy of Helium-4 and Alpha Decay Energy**

Given atomic masses: $^4_2$He = 4.002603 u, $^1_1$H = 1.007825 u, m_n = 1.008665 u, $^{238}_{92}$U = 238.050788 u, $^{234}_{90}$Th = 234.043601 u.

1. Calculate the binding energy per nucleon of $^4_2$He.
2. Calculate the Q-value for the α-decay: $^{238}_{92}\text{U} \longrightarrow {}^{234}_{90}\text{Th} + {}^4_2\text{He}$.
3. Determine the kinetic energy of the α particle (T_α) assuming the Uranium nucleus was at rest.

SOLUTION

Detailed Solution

1. Binding Energy of $^4_2$He: Constituents: 2 protons + 2 neutrons ≡ 2 Hydrogen atoms + 2 neutrons. Mass of constituents = 2(1.007825) + 2(1.008665) = 2.015650 + 2.017330 = 4.032980 u. Mass of ^{4}He atom = 4.002603 u. Mass defect Δm = 4.032980 - 4.002603 = 0.030377 u. E_b = 0.030377×931.5 MeV = 28.30 MeV. E_b/A = 28.30/4 = 7.07 MeV/nucleon. (Note: Very high for light nucleus, explains α stability).

2. Q-value for ^{238}U α-decay: $Q = [M(^{238}\text{U}) - M(^{234}\text{Th}) - M(^4\text{He})]c^2$. Δm = 238.050788 - 234.043601 - 4.002603 = 0.004584 u. Q = 0.004584 × 931.5 = 4.270 MeV. (Total kinetic energy shared by α and Th).

3. Kinetic Energy of α particle (T_α): Conservation of momentum: $p_\alpha = p_{\text{Th}}$ (magnitudes equal, opposite directions). $T = p^2/2m \Rightarrow T \propto 1/m$. $T_\alpha/T_{\text{Th}} = m_{\text{Th}}/m_\alpha \approx 234/4 = 58.5$. $Q = T_\alpha + T_{\text{Th}} = T_\alpha(1 + m_\alpha/m_{\text{Th}}) = T_\alpha(1 + 4/234) = T_\alpha(238/234)$. $T_\alpha = Q \times \frac{234}{238}$ = 4.270 × 0.9832 = 4.198 MeV. $T_{\text{Th}} = Q - T_\alpha$ = 0.072 MeV. Check: Typical α energy for ^{238}U is 4.20 MeV. Matches.

8 TYPES OF RADIATION, PROPERTIES AND APPLICATIONS

PROPERTY

Modes of Radioactive Decay

An unstable nucleus decays to reach a more stable configuration. Conservation laws: A, Z, energy-momentum, charge, lepton number.

- α decay: ${}^{A}_{Z}X \longrightarrow {}^{A-4}_{Z-2}Y + {}^{4}_{2}He$. Occurs in very heavy nuclei ($Z > 82$). Discrete energy spectrum.
- β^- decay: ${}^{A}_{Z}X \rightarrow {}^{A}_{Z+1}Y + {}^{0}_{-1}e + \bar{\nu}_e$. Neutron-rich. $n \rightarrow p + e^- + \bar{\nu}_e$. Continuous energy spectrum (shared with neutrino).
- β^+ decay: ${}^{A}_{Z}X \rightarrow {}^{A}_{Z-1}Y + {}^{0}_{+1}e + \nu_e$. Proton-rich. $p \rightarrow n + e^+ + \nu_e$. Requires $Q > 2m_ec^2 = 1.022$ MeV.
- Electron Capture (EC): ${}^{A}_{Z}X + {}^{0}_{-1}e \rightarrow {}^{A}_{Z-1}Y + \nu_e$. Inner shell electron captured. Alternative to β^+ if $Q < 1.022$ MeV.
- γ decay: ${}^{A}_{Z}X^* \rightarrow {}^{A}_{Z}X + \gamma$. Excited nucleus de-excites. No change in A or Z. Often follows α or β decay leaving daughter in metastable state (e.g., ${}^{99m}Tc$).

KEY POINTS

Properties of Radiations

Property	α particle	β^- particle	γ ray
Nature	${}^{4}_{2}He$ nucleus	Electron (${}^{0}_{-1}e$)	High-energy photon
Charge	$+2e$	$-e$	0
Mass	≈ 4 u (6.64×10^{-27} kg)	$m_e = 9.11 \times 10^{-31}$ kg	0 (energy $E = h\nu$)
Speed	~ $0.05c$ - $0.1c$	Up to ~ $0.99c$ (relativistic)	c
Ionising Power	Very High (dense track)	Medium	Low (indirect)
Range in Air	Few cm ($\approx$ 5 - 8 cm)	~ 1 m/MeV	Very large (km)
Penetration	Stopped by paper / skin	Stopped by few mm Al / perspex	Requires thick Pb / concrete
Deflection in E-field	Towards negative plate (small)	Towards positive plate (large)	None
Deflection in B-field	Perpendicular to $\vec{v}, \vec{B}$ (LH rule)	Opposite direction to α (tighter)	None

METHOD **Identifying Radiation using Electric and Magnetic Fields**

1. Setup: Source → narrow beam (collimator) → uniform field region → detector (photographic plate, GM tube, or fluorescent screen). Vacuum or low pressure for α/β.
2. Electric Field ($\vec{E}$): Force $\vec{F} = q\vec{E}$. Parabolic path if v constant.
   - α ($q = +2e$): Deflected towards negative plate, small curvature (large mass $m \approx 4u$).
   - β^- ($q = -e$): Deflected towards positive plate, large curvature (small mass m_e).
   - γ ($q = 0$): No deflection, straight line.
3. Magnetic Field ($\vec{B}$, into page ⊗): Force $\vec{F} = q(\vec{v} \wedge \vec{B})$. Circular path $r = mv/(qB)$. Use Fleming's Left Hand Rule (Motor Rule) for positive charge; reverse for negative.
   - α: Deflected upwards (if beam right, B into page), radius $r_\alpha = \frac{m_\alpha v}{2eB}$.
   - β^-: Deflected downwards, radius $r_\beta = \frac{m_e v}{eB} \ll r_\alpha$ (for same v). Relativistic correction often needed for β: $r = \frac{\gamma m v}{qB}$.
   - γ: No deflection.
4. Key distinction: α and β deflect in opposite directions in both fields. γ never deflects. β deflection $\gg$ α deflection for same energy (or same velocity).

WORKED EXAMPLE 9 **Deflection in a Magnetic Field**

A mixed beam of α particles and β^- particles, both with kinetic energy 5.00 MeV, enters a uniform magnetic field of flux density 0.200 T perpendicular to their velocity.

1. Calculate the velocity of the α particle (mass $m_\alpha = 6.64 \times 10^{-27}$ kg) and the β particle (mass $m_e = 9.11 \times 10^{-31}$ kg). Note: Check if relativistic correction is needed for the β particle.
2. Determine the radius of curvature for each particle in the magnetic field.
3. Sketch the paths on a diagram (beam entering horizontally, $\vec{B}$ into page).

SOLUTION

Detailed Solution

1. Velocities: Kinetic energy $K = 5.00\ \text{MeV} = 5.00 \times 10^6 \times 1.60 \times 10^{-19} = 8.00e-13\ \text{J}$.

α particle (Non-relativistic check): $K = \frac{1}{2}m_\alpha v_\alpha^2 \Rightarrow v_\alpha = \sqrt{\frac{2K}{m_\alpha}} = \sqrt{\frac{2\times8.00\times10^{-13}}{6.64\times10^{-27}}} = \sqrt{2.41 \times 10^{14}} = 1.55e7\ \text{m s}^{-1}$. $v_\alpha/c = 1.55 \times 10^7/3.00 \times 10^8 = 0.052$ (5.2%c). Non-relativistic approximation valid.

β particle (Relativistic check): Classical $v = \sqrt{2K/m_e} = \sqrt{2 \times 8.00 \times 10^{-13}/9.11 \times 10^{-31}} =$

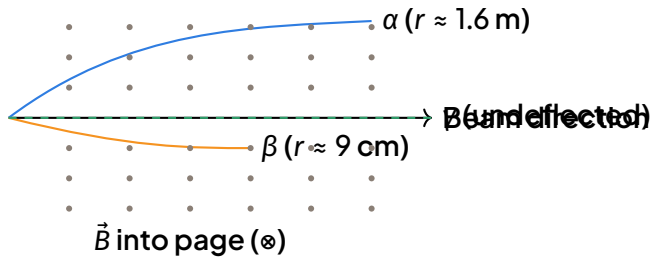
$1.33e9\ \text{m s}^{-1} > c$. Impossible. Must use relativity. Total Energy $E = K + m_0c^2 = 5.00\ \text{MeV} + 0.511\ \text{MeV} = 5.511\ \text{MeV}$. $E = \gamma m_0c^2 \Rightarrow \gamma = \frac{5.511}{0.511} = 10.78$. $\gamma = \frac{1}{\sqrt{1-v^2/c^2}} \Rightarrow v = c\sqrt{1-1/\gamma^2} = 3.00 \times 10^8\sqrt{1-1/116.2} = 2.987e8\ \text{m s}^{-1} \approx 0.996c$.

2. Radius of curvature $r = \frac{p}{qB} = \frac{\gamma mv}{qB}$ (Relativistic momentum).

α particle: $q_\alpha = +2e = 3.20 \times 10^{-19}$ C. $p_\alpha = m_\alpha v_\alpha = 6.64 \times 10^{-27} \times 1.55 \times 10^7 = 1.03 \times 10^{-19}$ kg m/s. $r_\alpha = \frac{1.03\times10^{-19}}{3.20\times10^{-19}\times0.200} = 1.61$ m.

β particle: $q_\beta = -e = 1.60 \times 10^{-19}$ C. Relativistic momentum $p_\beta = \frac{1}{c}\sqrt{E^2-(m_0c^2)^2}$. $E = 5.511\ \text{MeV} = 8.82 \times 10^{-13}$ J. $m_0c^2 = 0.511\ \text{MeV} = 8.18 \times 10^{-14}$ J. $p_\beta c = \sqrt{(8.82 \times 10^{-13})^2 - (8.18 \times 10^{-14})^2} \approx 8.78 \times 10^{-13}$ J. $p_\beta = 2.93 \times 10^{-21}$ kg m/s. $r_\beta = \frac{2.93\times10^{-21}}{1.60\times10^{-19}\times0.200} = 0.0915$ m = 9.15 cm.

3. Sketch Description:



Paths in magnetic field (qualitative sketch). α curves slightly up (large r). β curves sharply down (small r). γ straight.

Note: α deflects up (Fleming LH rule: current = velocity, field = into page → force up). β has negative charge → force down.

DID YOU KNOW? **Applications in Cameroon Context**

- Radiotherapy: ^{60}Co (gamma, 1.17 MeV & 1.33 MeV) units in Yaoundé and Douala hospitals for cancer treatment. Being replaced by Linear Accelerators (Linacs).
- Industrial Gauges: β sources (e.g., ^{90}Sr) for paper/plastic thickness control in factories (e.g., paper mill in Edéa). γ sources (^{137}Cs, ^{60}Co) for level gauges in oil tanks (SONARA refinery, Limbe) and density gauges.
- Tracers: ^{131}I (thyroid diagnosis), ^{99m}Tc (SPECT imaging - see worked example later). ^{3}H, ^{14}C in hydrology (tracing groundwater in Sahel regions).
- Smoke Detectors: ^{241}Am (α source) ionises air; smoke reduces current → alarm.
- Sterilisation: Gamma irradiation (^{60}Co) of medical equipment, food preservation (potatoes, spices).

9 MASS DEFECT AND NUCLEAR ENERGY

PROPERTY **Radioactive Decay Law**

The decay of a radioactive sample is a random process at the level of single nuclei, but statistically predictable for large numbers.

- Decay Constant λ: Probability per unit time that a given nucleus will decay. Unit: s^{-1}.
- Activity A: Rate of decay $A = -\frac{dN}{dt} = \lambda N$. Unit: Becquerel (Bq = s^{-1}). Old unit: Curie (Ci = 3.7e10 Bq).
- Differential Equation: $\frac{dN}{dt} = -\lambda N$.
- Solution: $N(t) = N_0 e^{-\lambda t}$ and $A(t) = A_0 e^{-\lambda t}$.
- Half-Life $T_{1/2}$: Time for half the nuclei to decay. $N(T_{1/2}) = N_0/2$.

$$T_{1/2} = \frac{\ln 2}{\lambda} \approx \frac{0.693}{\lambda}$$

- Fraction remaining after time t: $\frac{N}{N_0} = e^{-\lambda t} = \left(\frac{1}{2}\right)^{t/T_{1/2}}$.

METHOD **Radioactive Decay Law and Half-Life Calculations**

1. Identify the knowns: Initial number of nuclei N_0 (or initial activity A_0), half-life $T_{1/2}$ (or decay constant λ), and time elapsed t.
2. Convert half-life to decay constant: $\lambda = \frac{\ln 2}{T_{1/2}}$. Ensure consistent time units (seconds, years, etc.).
3. Apply the decay law: Number of remaining nuclei $N = N_0 e^{-\lambda t}$. Activity $A = A_0 e^{-\lambda t} = \lambda N$.
4. Calculate fraction remaining/decayed: Fraction remaining = $e^{-\lambda t} = \left(\frac{1}{2}\right)^{t/T_{1/2}}$. Fraction decayed = $1 - e^{-\lambda t}$.
5. Check units: Activity in Becquerel (Bq = s^{-1}), time in seconds (or consistent unit), λ in s^{-1}.
6. Reflex: For integer numbers of half-lives ($t = nT_{1/2}$), use $N = N_0/2^n$ directly for speed.

WORKED EXAMPLE 10 **Activity of a Cobalt-60 Source**

A radiotherapy unit in a Yaoundé hospital uses a $^{60}_{27}\mathrm{Co}$ source with an initial activity of $3.7e12$ Bq (100 Ci). The half-life of $^{60}\mathrm{Co}$ is 5.27 years.

1. Calculate the decay constant λ in s^{-1}.
2. Determine the activity after 3.00 years.
3. After how long will the activity fall to $1.00e11$ Bq?

SOLUTION

Detailed Solution

1. Decay constant λ: Convert half-life to seconds: $T_{1/2} = 5.27\ \text{years} \times 365.25\ \frac{\text{days}}{\text{year}} \times 24\ \frac{\text{h}}{\text{day}} \times 3600\ \frac{\text{s}}{\text{h}} = 1.663 \times 10^{8}\ \text{s}$.

$$\lambda = \frac{\ln 2}{T_{1/2}} = \frac{0.693}{1.663 \times 10^{8}\ \text{s}} = 4.17e-9\ \text{s}^{-1}$$

2. Activity after $t = 3.00$ years: Convert t to seconds: $t = 3.00 \times 3.15576 \times 10^{7}\ \text{s} = 9.467 \times 10^{7}\ \text{s}$. Use $A = A_0 e^{-\lambda t}$:

$$A = 3.7 \times 10^{12} \times \exp\left(-4.17 \times 10^{-9} \times 9.467 \times 10^{7}\right)$$

$$\lambda t = 0.395 \quad \Rightarrow \quad e^{-0.395} = 0.674$$

$$A = 3.7 \times 10^{12} \times 0.674 = 2.49e12\ \text{Bq}$$

Alternative (using years directly): $\lambda_y = \ln 2 / 5.27 = 0.1315\ \text{year}^{-1}$. $A = 3.7 \times 10^{12} \times e^{-0.1315 \times 3} = 2.49e12$ Bq.

3. Time for $A = 1.00e11$ Bq:

$$\frac{A}{A_0} = e^{-\lambda t} \quad \Rightarrow \quad \frac{1.00 \times 10^{11}}{3.7 \times 10^{12}} = 0.0270 = e^{-\lambda t}$$

$$-\lambda t = \ln(0.0270) = -3.612 \quad \Rightarrow \quad t = \frac{3.612}{\lambda}$$

Using $\lambda = 0.1315\ \text{year}^{-1}$:

$$t = \frac{3.612}{0.1315} = 27.5\ \text{years}$$

Answer: $\lambda = 4.17e-9\ \text{s}^{-1}$; $A(3\ \text{yr}) = 2.49e12$ Bq; $t = 27.5$ years.

METHOD **Balancing Nuclear Equations and Decay Series**

1. Conservation Laws: Total nucleon number A (mass number) and total proton number Z (atomic number) are conserved on both sides of the arrow.

2. Identify the radiation:
   - α-decay: $A \to A-4, Z \to Z-2$ (emits ${}^{4}_{2}\mathrm{He}$).
   - β^--decay: A constant, $Z \to Z+1$ (emits ${}^{0}_{-1}\mathrm{e} + \bar{\nu}_e$).
   - β^+-decay / EC: A constant, $Z \to Z-1$ (emits ${}^{0}_{+1}\mathrm{e} + \nu_e$ or captures e^-).
   - γ-decay: A and Z unchanged (energy release only).
3. Step-by-step for series: Write the parent nucleus. Apply decay 1 → daughter 1. Apply decay 2 → daughter 2, etc. Verify final stable isotope.
4. Notation: Use ${}^{A}_{Z}\mathrm{X}$ format. For electrons/positrons: ${}^{0}_{-1}\mathrm{e}$, ${}^{0}_{+1}\mathrm{e}$. For neutrinos: Ξ_e, $\bar{\Xi}_e$.

WORKED EXAMPLE 11 **Natural Radioactive Series: Uranium-238**

The ${}^{238}_{92}\mathrm{U}$ series ends at stable ${}^{206}_{82}\mathrm{Pb}$. The sequence of decays is: $\alpha, \beta, \beta, \alpha, \alpha, \alpha, \alpha, \beta, \beta, \alpha, \beta, \alpha, \beta$.

1. Write the full balanced nuclear equation for the first three decays.
2. Determine the total number of α and β particles emitted in the complete series.
3. Identify the noble gas isotope produced in this series (Radon) and its half-life (3.82 days). Why is it a radiological hazard in homes?

SOLUTION

Detailed Solution

1. First three decays: Start: ${}^{238}_{92}\mathrm{U}$
   - Decay 1 (α): $A = 234, Z = 90 \to {}^{234}_{90}\mathrm{Th}$ ${}^{238}_{92}\mathrm{U} \longrightarrow {}^{234}_{90}\mathrm{Th} + {}^{4}_{2}\mathrm{He}$
   - Decay 2 (β^-): $A = 234, Z = 91 \to {}^{234}_{91}\mathrm{Pa}$ ${}^{234}_{90}\mathrm{Th} \to {}^{234}_{91}\mathrm{Pa} + {}^{0}_{-1}\mathrm{e} + \bar{\Xi}_e$
   - Decay 3 (β^-): $A = 234, Z = 92 \to {}^{234}_{92}\mathrm{U}$ ${}^{234}_{91}\mathrm{Pa} \to {}^{234}_{92}\mathrm{U} + {}^{0}_{-1}\mathrm{e} + \bar{\Xi}_e$

2. Total α and β count: Let n_α = number of α, n_β = number of β^-. Change in A: $\Delta A = 238-206 = 32$. Only α changes A (by -4). $4n_\alpha = 32 \Rightarrow n_\alpha = 8$. Change in Z: $\Delta Z = 92 - 82 = 10$. α decreases Z by 2, β^- increases Z by 1. $-2n_\alpha + n_\beta = -10 \Rightarrow -2(8) + n_\beta = -10 \Rightarrow n_\beta = 6$. Check: The given sequence has 8 α and 6 β (count them: $\alpha\beta\beta\alpha\alpha\alpha\beta\beta\alpha\beta\alpha\beta$). Matches.

3. Radon isotope: Look at the series steps. After 6 decays ($\alpha\beta\beta\alpha\alpha\alpha$), we reach $A = 222, Z = 86$ (Radon). ${}^{226}_{88}\mathrm{Ra} \to [\Xi]{}^{222}_{86}\mathrm{Rn}$ (Radon-222). Half-life $T_{1/2} = 3.82$ days. Hazard: ${}^{222}\mathrm{Rn}$ is a noble gas, chemically inert. It diffuses from soil/rocks (granite areas in West Cameroon, Adamawa) into basements and poorly ventilated rooms. Being a gas, it is inhaled. Its short-lived solid daughters (${}^{218}\mathrm{Po}$, ${}^{214}\mathrm{Pb}$, ${}^{214}\mathrm{Bi}$, ${}^{214}\mathrm{Po}$) attach to dust aerosols, deposit in lungs, and emit high-energy α particles, causing significant radiation dose to bronchial epithelium → lung cancer risk. It is the second leading cause of lung cancer after smok-

ing.

METHOD **Radioactive Dating using Carbon-14**

1. Principle: Living organisms maintain equilibrium $^{14}C/^{12}C$ ratio ($\approx 1.3 \times 10^{-12}$) via $^{14}N(n,p)^{14}C$ in atmosphere and photosynthesis/feeding. At death, intake stops. ^{14}C decays (β^-, $T_{1/2} = 5730$ yr). Ratio decreases.
2. Formula: $\frac{A}{A_0} = \frac{N}{N_0} = e^{-\lambda t} = \left(\frac{1}{2}\right)^{t/T_{1/2}}$.
3. Solve for age t: $t = -\frac{1}{\lambda}\ln\left(\frac{A}{A_0}\right) = -T_{1/2}\frac{\ln(A/A_0)}{\ln 2}$.
4. Assumptions/Limits: Constant atmospheric ^{14}C production (calibration curves needed for precision), closed system (no contamination), range $\approx$ 100 - 50,000 years (beyond: activity too low; below: not enough decay).
5. Units: $T_{1/2} = 5730$ yr (Libby half-life) or 5700 yr (Cambridge). Use value given in question.

WORKED EXAMPLE 12 **Age of a Wooden Artefact**

A wooden mask from a Cameroonian museum collection shows a ^{14}C activity of 0.12 Bqpergramofcarbon. The activity of living wood is 0.25 Bq/g. The half-life of ^{14}C is taken as 5730 years.

1. Calculate the age of the mask.
2. The measured activity has an uncertainty of ±0.01 Bq/g. Determine the uncertainty in the age.
3. Why can't this method date a dinosaur bone (65 millionyearsold)?

SOLUTION

Detailed Solution

1. Age calculation: $\frac{A}{A_0} = \frac{0.12}{0.25} = 0.48$. $\lambda = \frac{\ln 2}{5730} = 1.2097 \times 10^{-4}$ yr^{-1}. $t = -\frac{1}{\lambda}\ln(0.48) = -\frac{1}{1.2097\times10^{-4}} \times (-0.7340) = 6068$ years. Using half-life formula: $t = -5730 \times \frac{\ln 0.48}{\ln 2} = 5730 \times 1.058 = 6062$ years. Age $\approx$ 6070 years (Neolithic period).
2. Uncertainty propagation: $A = 0.12 \pm 0.01 \Rightarrow A_{min} = 0.11, A_{max} = 0.13$. $t_{max} = -5730\frac{\ln(0.11/0.25)}{\ln 2} = -5730\frac{\ln 0.44}{\ln 2} = 6600$ yr. $t_{min} = -5730\frac{\ln(0.13/0.25)}{\ln 2} = -5730\frac{\ln 0.52}{\ln 2} = 5580$ yr. Uncertainty $\Delta t \approx \frac{6600-5580}{2} = 510$ years. Or via derivative: $\Delta t = \frac{T_{1/2}}{\ln 2}\frac{\Delta A}{A} = \frac{5730}{0.693} \times \frac{0.01}{0.12} = 8268 \times 0.0833 = 689$ years. (Asymmetric errors due to log; range method better). Age

= 6070 ± 510 years.
3. Dinosaur bone limit: $t = 65 \times 10^6$ yr. Number of half-lives $n = 65 \times 10^6 / 5730 \approx 11340$. Fraction remaining = $(1/2)^{11340} \approx 10^{-3414}$. Effectively zero. Activity would be immeasurably small (far below background radiation). Carbon-14 dating is limited to ≈ 8 - 10 half-lives (~ 50,000 years). For millions of years, use $^{40}K/^{40}Ar$ ($T_{1/2} = 1.25 \times 10^9$ yr) or $^{238}U/^{206}Pb$ on surrounding volcanic rock (zircon crystals).

10 NUCLEAR FISSION AND NUCLEAR FUSION

PROPERTY **Nuclear Fission**

Discovery: Hahn, Strassmann, Meitner, Frisch (1938–39). Neutron-induced fission of ^{235}U. Typical Reaction:

$$^{235}_{92}U + ^{1}_{0}n \rightarrow ^{A_1}_{Z_1}X + ^{A_2}_{Z_2}Y + k\,^{1}_{0}n + Q$$

Conservation: $A_U + 1 = A_1 + A_2 + k$; $Z_U = Z_1 + Z_2$. Fragments X, Y are neutron-rich → undergo β^- decay chains (fission products). Energy Q: $Q = [M(^{235}U) + m_n - M(X) - M(Y) - km_n]c^2$. Or using Binding Energy: $Q = E_b(X) + E_b(Y) - E_b(^{235}U)$. (Neutron has $E_b = 0$). Typical values: $Q \approx 200$ MeV per fission. Distribution: Kinetic energy of fragments (~ 165 MeV), prompt γ (~ 7 MeV), kinetic energy of neutrons (~ 5 MeV), β/γ from fission products (~ 23 MeV), neutrinos (~ 10 MeV, lost). Chain Reaction: Requires $k \geq 1$ neutron per fission to sustain. Critical mass, moderator (slow neutrons increase σ_f for ^{235}U), control rods (absorb neutrons: B, Cd, Hf).

METHOD **Energy Released in Nuclear Fission**

1. Write balanced equation: Check A and Z conservation. Identify fragments and number of neutrons k.
2. Calculate Q-value: Use atomic masses. $Q = (\sum m_{\text{initial}} - \sum m_{\text{final}})c^2$. 1 u = 931.5 MeV.
3. Power calculation: $P = \frac{\text{Energy}}{\text{time}} = R \times Q$, where R is fission rate (fissions/s). Mass consumption rate $\dot{m} = R \times m_{\text{nucleus}}$.

WORKED EXAMPLE 13 **Fission of Uranium-235**

A thermal neutron induces fission in $^{235}_{92}$U producing $^{141}_{56}$Ba, $^{92}_{36}$Kr and 3 neutrons. Atomic masses: ^{235}U = 235.043930 u, ^{141}Ba = 140.914411 u, ^{92}Kr = 91.926156 u, neutron = 1.008665 u.

1. Write the balanced nuclear equation.
2. Calculate the total energy released Q in MeV.
3. If a reactor core produces 3000 MW of thermal power, calculate the mass of ^{235}U consumed (fissioned) per day. (Assume 200 MeV/fission for usable energy).

SOLUTION

Detailed Solution
1. Balanced Equation: $^{235}_{92}\mathrm{U} + ^{1}_{0}\mathrm{n} \rightarrow ^{141}_{56}\mathrm{Ba} + ^{92}_{36}\mathrm{Kr} + 3\,^{1}_{0}\mathrm{n}$ Check: $A: 235 + 1 = 141 + 92 + 3 = 236$. $Z: 92 = 56 + 36 = 92$. OK.
2. Energy Released Q: Initial mass $m_i = M(^{235}\mathrm{U}) + m_n = 235.043930 + 1.008665 = 236.052595$ u. Final mass $m_f = M(^{141}\mathrm{Ba}) + M(^{92}\mathrm{Kr}) + 3m_n = 140.914411 + 91.926156 + 3(1.008665) = 232.840567 + 3.025995 = 235.866562$ u. Mass defect $\Delta m = m_i - m_f = 236.052595 - 235.866562 = 0.186033$ u. $Q = 0.186033 \times 931.5 = 173.3$ MeV. (Note: This specific channel yields ~ 173 MeV. Average over all channels is ~ 200 MeV. The difference is in excitation energy of fragments / prompt gammas).
3. Mass consumption rate: Power $P = 3000$ MW $= 3.00 \times 10^9$ J/s. Energy per fission $E_f = 200$ MeV $= 200 \times 1.60 \times 10^{-13} = 3.20 \times 10^{-11}$ J. Fissions per second $R = P/E_f = 3.00 \times 10^9 / 3.20 \times 10^{-11} = 9.375 \times 10^{19}$ s^{-1}. Seconds per day = 86400 s. Total fissions/day = $9.375 \times 10^{19} \times 86400 = 8.10 \times 10^{24}$. Mass of one ^{235}U atom = $235 \times 1.6605 \times 10^{-27} = 3.90 \times 10^{-25}$ kg. Mass consumed = $8.10 \times 10^{24} \times 3.90 \times 10^{-25} = 3.16$ kg/day. Result: ≈ 3.2 kg of ^{235}U fissioned per day. (Total fuel load much higher due to enrichment, non-fission capture, etc.).

PROPERTY **Nuclear Fusion**

Principle: Light nuclei combine to form a heavier, more tightly bound nucleus. Energy released because E_b/A increases. Conditions (Lawson Criterion):

- High Temperature ($T > 10^7$ K, ~ 1 keV) for kinetic energy to overcome Coulomb barrier ($V_C \approx \frac{Z_1 Z_2 e^2}{4\pi\epsilon_0 r}$). Quantum tunnelling essential.
- High Density/Pressure (n) for collision frequency.
- Confinement Time (τ_E): Energy must be retained long enough. Product $n\tau_E$ must exceed threshold (Lawson criterion).

Main Reactions:

- p-p Chain (Stars like Sun): $4\,^{1}_{1}\mathrm{H} \rightarrow ^{4}_{2}\mathrm{He} + 2\,^{0}_{+1}\mathrm{e} + 2\nu_e + \gamma$ ($Q \approx 26.7$ MeV). Slow (weak interaction).

- D-T (Tokamaks/ITER): ${}^{2}_{1}\mathrm{H} + {}^{3}_{1}\mathrm{H} \longrightarrow {}^{4}_{2}\mathrm{He} + {}^{1}_{0}\mathrm{n}$ (Q = 17.6 MeV). Largest cross-section at lowest temperature (~ 10 keV).
- D-D: ${}^{2}_{1}\mathrm{H} + {}^{2}_{1}\mathrm{H} \longrightarrow {}^{3}_{1}\mathrm{H} + {}^{1}_{1}\mathrm{H}$ (4.0 MeV) or ${}^{3}_{2}\mathrm{He} + {}^{1}_{0}\mathrm{n}$ (3.3 MeV).

Advantages: No long-lived high-level waste (activation of vessel only), abundant fuel (Deuterium in seawater ≈ 0.015%, Tritium bred from Lithium: ${}^{6}\mathrm{Li} + \mathrm{n} \longrightarrow {}^{4}\mathrm{He} + \mathrm{T}$), inherent safety (no chain reaction, meltdown impossible).

METHOD **Energy Released in Nuclear Fusion**

1. Write balanced equation: Check A, Z conservation.
2. Calculate Q-value: $Q = (\sum m_{\text{initial}} - \sum m_{\text{final}})c^2$. Use atomic masses (electrons balance if Z conserved).
3. Kinetic Energy Partition: For 2-body final state at rest, momentum conservation ⇒ $p_1 = p_2$. $T_1/T_2 = m_2/m_1$. $Q = T_1 + T_2$.
4. Fuel Energy Density: Moles of fuel → Number of reactions ×Q.

WORKED EXAMPLE 14 **Deuterium–Tritium Fusion Energy**

Consider the D-T fusion reaction: ${}^{2}_{1}\mathrm{H} + {}^{3}_{1}\mathrm{H} \longrightarrow {}^{4}_{2}\mathrm{He} + {}^{1}_{0}\mathrm{n}$. Atomic masses: ${}^{2}\mathrm{H}$ = 2.014102 u, ${}^{3}\mathrm{H}$ = 3.016049 u, ${}^{4}\mathrm{He}$ = 4.002603 u, neutron = 1.008665 u.

1. Verify the Q-value is 17.6 MeV.
2. Calculate the kinetic energy of the neutron and the alpha particle (Helium-4 nucleus) assuming the reactants are at rest.
3. 1 g of Deuterium fuses with 1.5 g of Tritium (stoichiometric). Calculate the energy released in Joules and compare to 1 tonne of coal (30 GJ).

SOLUTION

Detailed Solution

1. Q-value verification: Initial mass m_i = 2.014102 + 3.016049 = 5.030151 u. Final mass m_f = 4.002603 + 1.008665 = 5.011268 u. Δm = 5.030151 - 5.011268 = 0.018883 u. Q = 0.018883 × 931.5 = 17.59 MeV ≈ 17.6 MeV. Verified.

2. Kinetic Energy Partition: Momentum conservation: $p_n = p_\alpha$ (magnitudes). $T = p^2/2m$ ⇒ $T_n/T_\alpha = m_\alpha/m_n \approx 4/1 = 4$. $Q = T_n + T_\alpha = T_n + T_n/4 = 1.25T_n$. $T_n = Q/1.25 = 17.6/1.25 = 14.08$ MeV. $T_\alpha = Q - T_n = 17.6 - 14.08 = 3.52$ MeV. (Neutron carries 80% of energy → major challenge for material shielding and heat extraction in blankets).

3. Energy from 1g Deuterium + 1.5g Tritium: Molar mass D = 2 g/mol, T = 3 g/mol. Moles of D = 1/2 = 0.5 mol. Moles of T = 1.5/3 = 0.5 mol. Number of pairs $N = 0.5 \times N_A = 0.5 \times 6.022 \times 10^{23} = 3.011 \times 10^{23}$. Total Energy $E = N \times Q = 3.011 \times 10^{23} \times 17.6 \times 1.60 \times 10^{-13}$ J. $E = 3.011 \times 10^{23} \times 2.816 \times 10^{-12} = 8.48 \times 10^{11}$ J = 848 GJ. Comparison: 848 GJ/30 GJ/tonne ≈ 28.3 tonnes of coal. Conclusion: 2.5 g of fusion fuel ≈ 28 tonnes of coal. Energy density ~ 10^7 times chemical.

METHOD **Comprehensive Nuclear Physics Problem Solving**

1. Read carefully: Identify the process (decay, fission, fusion, dating, stability).
2. List Data: Extract masses, half-lives, energies, constants from text or data sheet.
3. Conservation Checks: Nucleon number A, Proton number Z, Energy-Momentum, Charge.
4. Select Formulas:
   - Decay: $N = N_0 e^{-\lambda t}$, $A = \lambda N$, $T_{1/2} = \ln 2/\lambda$.
   - Energy: $Q = \Delta m c^2$, $E_b = \Delta m c^2$, $T_\alpha = Q \frac{A_{\text{daughter}}}{A_{\text{daughter}}+4}$.
   - Dating: $t = -T_{1/2} \ln(A/A_0)/\ln 2$.
5. Unit Consistency: Convert all to SI (kg, s, J) or Nuclear (u, MeV) consistently. 1 u = 931.5 MeV/c^2 = 1.6605×10^{-27} kg.
6. Significant Figures: Match input data (usually 3 or 4 s.f.). State units clearly.
7. Physical Comment: Interpret result (e.g., "This isotope is stable", "Energy is released", "Age is within range").

WORKED EXAMPLE 15 **Analysis of a Medical Radioisotope: Technetium-99m**

Technetium-99m ($^{99m}_{43}$Tc, m = metastable) is the workhorse of nuclear medicine (SPECT imaging). It decays by isomeric transition (IT) to $^{99}_{43}$Tc (ground state) emitting a 140 keV γ ray. $T_{1/2}$ = 6.01 hours. The ground state ^{99}Tc decays by β^- to stable $^{99}_{44}$Ru with $T_{1/2}$ = 2.13e5 years. A ^{99}Mo/^{99m}Tc generator produces ^{99m}Tc via β^- decay of ^{99}Mo ($T_{1/2}$ = 66.0 hours). At t = 0 (elution), the generator yields 1.00 GBq of ^{99m}Tc.

1. Write the decay equations for ^{99}Mo, ^{99m}Tc (IT), and ^{99}Tc (β^-).
2. Calculate the activity of ^{99m}Tc remaining 12.0 hours after elution (ignore ^{99}Mo breakthrough).

3. Why is the 140 keV γ ray ideal for gamma camera imaging? (Consider penetration in tissue and collimator efficiency).

4. The ^{99}Tc daughter accumulates. Calculate its activity 24 hours after elution. Is it a significant radiation hazard to the patient? (Assume no biological clearance).

SOLUTION

Detailed Solution

1. Decay Equations: $^{99}_{42}\text{Mo} \rightarrow ^{99m}_{43}\text{Tc} + ^{0}_{-1}\text{e} + \bar{\nu}_e$ ($T_{1/2}$ = 66.0 h, β^- decay) $^{99m}_{43}\text{Tc} \rightarrow ^{99}_{43}\text{Tc} + \nu$ ($T_{1/2}$ = 6.01 h, Isomeric Transition, E_γ = 140 keV) $^{99}_{43}\text{Tc} \rightarrow ^{99}_{44}\text{Ru} + ^{0}_{-1}\text{e} + \bar{\nu}_e$ ($T_{1/2} = 2.13 \times 10^5$ yr, β^- decay, $E_{\beta,\text{max}}$ = 292 keV)

2. Activity of ^{99m}Tc at t = 12.0 h: $\lambda_{Tc} = \ln 2/6.01 = 0.1153\ \text{h}^{-1}$. $A = A_0 e^{-\lambda t} = 1.00 \times e^{-0.1153 \times 12.0} = 1.00 \times e^{-1.384} = 1.00 \times 0.250 = 0.250$ GBq = 250 MBq. (Exactly 2 half-lives: 12.0/6.01 ≈ 2.00 ⇒ 1/4 activity).

3. Suitability of 140 keV γ:

- Penetration: In soft tissue (water), linear attenuation coefficient $\mu \approx 0.15\ \text{cm}^{-1}$ at 140 keV. Half-value layer (HVL) ≈ 4.6 cm. Allows imaging of deep organs (heart, brain) with sufficient photons escaping the body.
- Collimation: Lead collimators (parallel hole) require septa thick enough to stop γ. For 140 keV, septa ≈ 0.2 - 0.3 mm Pb suffice (HVL in Pb ≈ 0.03 cm). Allows high geometric resolution (hole diameter ~ 1 - 2 mm). Higher energy (e.g., 511 keV) needs thick septa → poor resolution/sensitivity. Lower energy (e.g., 80 keV) absorbed in tissue → poor depth penetration.
- Detection: NaI(Tl) scintillators have peak efficiency ~ 80 - 90% at 140 keV.

Conclusion: Optimal compromise between tissue penetration and collimator/detector performance.

4. Activity of ^{99}Tc at t = 24 h: ^{99m}Tc decays to ^{99}Tc. This is a Parent (P) → Daughter (D) decay where $\lambda_P \gg \lambda_D$ ($T_{1/2,P}$ = 6 h ≪ $T_{1/2,D} = 2.13 \times 10^5$ yr). Activity of daughter $A_D(t) = A_P(0)\frac{\lambda_D}{\lambda_D - \lambda_P}(e^{-\lambda_P t} - e^{-\lambda_D t}) + A_D(0)e^{-\lambda_D t}$. Since $\lambda_D \approx 0$ (very long half-life), $e^{-\lambda_D t} \approx 1$, $\frac{\lambda_D}{\lambda_D - \lambda_P} \approx 0$. $A_D(t) \approx A_P(0)(1 - e^{-\lambda_P t})$. $A_P(0)$ = 1.00 GBq. $\lambda_P = 0.1153\ \text{h}^{-1}$. t = 24 h. $e^{-\lambda_P t} = e^{-0.1153 \times 24} = e^{-2.767} = 0.063$. $A_D(24\ \text{h}) \approx 1.00 \times (1 - 0.063) = 0.937$ GBq = 937 MBq. Hazard Assessment: ^{99}Tc emits low-energy β^- (E_{max} = 292 keV, E_{avg} ≈ 100 keV). Range in tissue ≈ 0.5 mm. No significant γ emission. Dose is highly localized. However, ^{99}Tc chemistry (pertechnetate TcO_4^-) concentrates in thyroid, stomach, salivary glands. Effective dose coefficient (ingestion) ≈ 6.4×10^{-10} Sv/Bq. Committed dose ≈ $9.37 \times 10^8 \times 6.4 \times 10^{-10} \approx 0.6$ mSv. This is low (background ~ 2.4 mSv/yr, CT scan ~ 10 mSv). Not a significant acute hazard, but contributes to long-term stochastic risk. Justifies use in diagnostics.

WARNING **Common Mistakes in Radioactivity**

- Atomic vs Nuclear Masses: Always use atomic masses from data tables for Q-value calculations. They include electrons which cancel correctly if Z is conserved. Using nuclear masses requires manual electron accounting.
- Relativistic β particles: Never use $K = \frac{1}{2}mv^2$ for β particles above ~ 0.1 MeV. Use $E = \gamma m_0 c^2$, $p = \gamma m_0 v$.
- Half-life units: Ensure λ and t have consistent time units (both seconds, or both years). $\lambda = \ln 2 / T_{1/2}$.
- Decay Series Counting: For α count: $\Delta A/4$. For β count: Solve $-2n_\alpha + n_\beta = \Delta Z$. Don't just count arrows in a diagram if not given.
- Activity vs Number of Nuclei: $A = \lambda N$. Don't confuse A (Bq) with N (dimensionless count).
- Carbon Dating Limit: $\approx 50,000$ years (~ 8 - 10 half-lives). Beyond that, use other isotopes (^{40}K, ^{238}U).

EXERCISE 6 **GCE Style Practice Questions**

- A sample of radioactive isotope has an initial activity of 800 Bq. After 12 hours, the activity is 100 Bq. Calculate the half-life of the isotope.
- Write the nuclear equation for the β^+ decay of $^{22}_{11}$Na. The daughter nucleus is $^{22}_{10}$Ne. Calculate the maximum kinetic energy of the positron given: $M(^{22}\text{Na}) = 21.994437$ u, $M(^{22}\text{Ne}) = 21.991385$ u. (1 u = 931.5 MeV).
- In a fission reactor, 1.00 kg of ^{235}U undergoes fission. Assuming an average energy release of 200 MeV per fission, calculate the total energy released in Joules.
- Explain why high temperatures (of order 10^7 K) are necessary for nuclear fusion to occur. Explain the role of quantum tunnelling.
- A rock sample contains ^{238}U and ^{206}Pb in a molar ratio of 1:3. Assuming no initial ^{206}Pb and no leakage, calculate the age of the rock. ($T_{1/2}(^{238}\text{U}) = 4.47e9$ years).

SOLUTION **Exercise 1**

$A/A_0 = 100/800 = 1/8 = (1/2)^3$. So 3 half-lives have passed in 12 hours. $T_{1/2} = 12/3 = 4$ hours.

SOLUTION **Exercise 2**

Equation: ${}^{22}_{11}\text{Na} \rightarrow {}^{22}_{10}\text{Ne} + {}^{0}_{+1}\text{e} + \nu_e$. $Q = [M({}^{22}\text{Na}) - M({}^{22}\text{Ne}) - 2m_e]c^2$. (Atomic mass of Na includes 11 e^-, Ne includes 10 e^-. Positron emission requires creation of e^+ mass, equivalent to losing 2 electron masses from atomic mass balance). $\Delta m = 21.994437 - 21.991385 - 2(0.00054858) = 0.003051 - 0.001097 = 0.001954$ u. $Q = 0.001954 \times 931.5 = 1.82$ MeV. $K_{max}(e^+) \approx Q = 1.82$ MeV (neglecting recoil of heavy Ne nucleus).

SOLUTION **Exercise 3**

Number of atoms in 1 kg ${}^{235}\text{U}$: $N = \frac{1000\text{ g}}{235\text{ g/mol}} \times 6.022 \times 10^{23} = 2.56 \times 10^{24}$. Energy per fission $= 200 \times 1.60 \times 10^{-13} = 3.20 \times 10^{-11}$ J. Total Energy $= 2.56 \times 10^{24} \times 3.20 \times 10^{-11} = 8.19e13$ J.

SOLUTION **Exercise 4**

High temperature gives nuclei high kinetic energy ($E_k \approx k_B T$) to approach within range of strong force (~ 1-3 fm), overcoming Coulomb repulsion $V_C = \frac{Z_1 Z_2 e^2}{4\pi\epsilon_0 r}$. At $T \sim 10^7$ K, $k_B T \approx 1$ keV, while $V_C \sim 100 - 1000$ keV. Classical physics forbids fusion. Quantum Tunnelling: Wave nature of particles allows a finite probability to penetrate the classically forbidden barrier region. The Gamow factor gives the probability $\propto \exp(-2\pi Z_1 Z_2 e^2/(4\pi\epsilon_0 \hbar v))$. Without tunnelling, fusion rates in stars would be zero.

SOLUTION **Exercise 5**

Molar ratio ${}^{206}\text{Pb} : {}^{238}\text{U} = 3 : 1$. Let N_0 be initial U atoms. $N_U = N_0 e^{-\lambda t}$. $N_{Pb} = N_0 - N_U = N_0(1 - e^{-\lambda t})$. Ratio $N_{Pb}/N_U = (1 - e^{-\lambda t})/e^{-\lambda t} = e^{\lambda t} - 1 = 3$. $e^{\lambda t} = 4 \Rightarrow \lambda t = \ln 4 = 2\ln 2$. $t = 2\ln 2/\lambda = 2T_{1/2} = 2 \times 4.47 \times 10^9 = 8.94e9$ years.

11 EXERCISES

11.1 I check my knowledge

EXERCISE 7 **Multiple choice questions**

1. Which of the following radiations has the highest ionising power?
   (a) α-particles
   (b) β-particles
   (c) γ-rays
   (d) X-rays
2. A radioactive nucleus ${}^{A}_{Z}X$ decays by emitting an α-particle. The resulting nucleus is:
   (a) ${}^{A-4}_{Z-2}Y$
   (b) ${}^{A-4}_{Z+2}Y$
   (c) ${}^{A}_{Z-1}Y$
   (d) ${}^{A}_{Z+1}Y$
3. The half-life of a radioactive isotope is 10 days. What fraction of the original sample remains after 30 days?
   (a) $\frac{1}{2}$
   (b) $\frac{1}{4}$
   (c) $\frac{1}{8}$
   (d) $\frac{1}{16}$
4. In a nuclear fission reaction, the total mass of the products is:
   (a) greater than the total mass of the reactants
   (b) equal to the total mass of the reactants
   (c) less than the total mass of the reactants
   (d) zero

EXERCISE 8 **True or False with justification**

State whether each statement is True or False and justify your answer briefly.

1. γ-rays are deflected by a magnetic field.
2. The mass defect of a nucleus is the difference between the mass of the nucleus and the sum of the masses of its constituent nucleons.
3. In β^- decay, the atomic number Z decreases by 1 while the mass number A remains constant.
4. Nuclear fusion requires very high temperatures to overcome the electrostatic repulsion between nuclei.

EXERCISE 9 **Definitions and direct calculations**

1. Define the following terms: (a) Half-life ($T_{1/2}$), (b) Activity (A), (c) Binding energy per nucleon.
2. The decay constant of a radioactive substance is $\lambda = 0.02\ \text{s}^{-1}$. Calculate its half-life.
3. A sample has an initial activity of 800 Bq. After 12 hours, its activity is 100 Bq. Determine the half-life of the sample.
4. Write the balanced nuclear equation for the α-decay of $^{226}_{88}\text{Ra}$ and the β^--decay of $^{14}_{6}\text{C}$.

EXERCISE 10 **Radiation identification and properties**

1. A radiation from a radioactive source is stopped by a sheet of paper and is strongly deflected by a magnetic field in a direction opposite to that of β-particles. Identify the radiation and state its charge.
2. Complete the following nuclear equations and identify the type of decay in each case:

$$^{238}_{92}\text{U} \longrightarrow {}^{234}_{90}\text{Th} + {}^{4}_{2}\text{He}$$
$$^{14}_{6}\text{C} \longrightarrow {}^{14}_{7}\text{N} + {}^{0}_{-1}\text{e}$$
$$^{214}_{83}\text{Bi} \longrightarrow {}^{214}_{84}\text{Po} + {}^{0}_{-1}\text{e}$$

3. For each radiation identified in (ii), state two distinct properties.

11.2 I practise

EXERCISE 11 Activity and decay law

A radioactive source used in a hospital has a half-life of 6 hours. At 8:00 am, its activity is measured as 1.6*e*6 Bq.

1. Calculate the decay constant λ in s^{-1}.
2. What will be the activity at 2:00 pm on the same day?
3. How long will it take for the activity to fall to 100000 Bq?
4. Sketch a graph of $\ln(A)$ against time t, indicating the slope and intercept.

EXERCISE 12 Mass defect and binding energy

Given the following masses: $m_p = 1.007276\,\text{u}$, $m_n = 1.008665\,\text{u}$, $m(^4_2\text{He}) = 4.002603\,\text{u}$, $1\,\text{u} = 931.5\,\text{MeV}/c^2$.

1. Calculate the mass defect of the helium-4 nucleus in u and in kg.
2. Calculate the total binding energy of the nucleus in MeV and in joules.
3. Determine the binding energy per nucleon for helium-4.
4. Comment on the stability of helium-4 relative to lighter nuclei using the binding energy per nucleon curve.

EXERCISE 13 Nuclear fission energy calculation

Consider the fission reaction induced by a thermal neutron:

$$^{235}_{92}\text{U} + ^{1}_{0}\text{n} \longrightarrow ^{94}_{38}\text{Sr} + ^{140}_{54}\text{Xe} + 2\,^{1}_{0}\text{n}$$

Atomic masses: $m(^{235}\text{U}) = 235.04393\,\text{u}$, $m(^{94}\text{Sr}) = 93.91536\,\text{u}$, $m(^{140}\text{Xe}) = 139.92164\,\text{u}$, $m_n = 1.008665\,\text{u}$.

1. Verify that the nucleon number and proton number are conserved.
2. Calculate the mass defect Δm for this reaction in u.
3. Calculate the energy released per fission in MeV and in joules.
4. Calculate the energy released by the fission of 1 g of pure ^{235}U.
5. Compare this energy with the heat of combustion of 1 g of petrol ($\approx 47\,\text{MJ}$). How many grams of petrol would be equivalent to 1 g of ^{235}U?

EXERCISE 14 **Carbon-14 dating**

The half-life of ^{14}C is 5730 years. A piece of wood from an archaeological site shows an activity of 25% of that found in living wood.

1. Explain the principle of carbon-14 dating, mentioning the role of cosmic rays and the carbon cycle.
2. Calculate the age of the piece of wood.
3. If the measurement of activity has an uncertainty of ±2%, estimate the uncertainty in the calculated age.
4. Why is carbon-14 dating not suitable for samples older than about 50,000 years?

11.3 I prepare for the GCE

EXERCISE 15 **Structured essay: Nuclear Reactors and Energy in Cameroon] [20 marks**

1. Draw a labelled diagram of a typical thermal nuclear reactor (e.g., Pressurised Water Reactor) showing the following components: reactor core, fuel rods, moderator, control rods, coolant, heat exchanger (steam generator), turbine, generator, condenser, cooling tower. [6 marks]
2. Explain the function of the moderator and the control rods in the reactor core. Why is it necessary to slow down neutrons? [4 marks]
3. Describe the process of nuclear fission of ^{235}U initiated by a thermal neutron, including the concept of a chain reaction and critical mass. [4 marks]
4. Discuss two advantages and two risks/disadvantages of generating electricity from nuclear power specifically in the context of Cameroon's energy needs and environment. [6 marks]

EXERCISE 16 **Binding energy curve, fission and fusion] [15 marks**

1. Sketch the binding energy per nucleon (B/A) against nucleon number (A) curve (Aston curve). Indicate the approximate position of iron-56 (^{56}Fe), uranium-235 (^{235}U), and helium-4 (^{4}He). [4 marks]
2. Using the curve, explain qualitatively why:

   (a) Energy is released when ^{235}U undergoes fission into two medium-mass nuclei.

(b) Energy is released when two light nuclei (e.g., deuterium and tritium) undergo fusion.

[4 marks]

3. For the fusion reaction ${}_{1}^{2}\mathrm{H} + {}_{1}^{3}\mathrm{H} \longrightarrow {}_{2}^{4}\mathrm{He} + {}_{0}^{1}\mathrm{n}$, the masses are: $m({}^{2}\mathrm{H}) = 2.014102\,\mathrm{u}$, $m({}^{3}\mathrm{H}) = 3.016049\,\mathrm{u}$, $m({}^{4}\mathrm{He}) = 4.002603\,\mathrm{u}$, $m_n = 1.008665\,\mathrm{u}$. Calculate the energy released in this reaction in MeV. [4 marks]

4. State two major difficulties in achieving controlled nuclear fusion on Earth for power generation. [3 marks]

EXERCISE 17 **Comprehensive radioactivity problem] [15 marks**

A radioactive series starts with nuclide ${}_{92}^{238}\mathrm{U}$ and ends with stable ${}_{82}^{206}\mathrm{Pb}$. The series involves a sequence of α and β^- decays.

1. Determine the total number of α particles and β^- particles emitted in the complete decay of one atom of ${}^{238}\mathrm{U}$ to ${}^{206}\mathrm{Pb}$. [3 marks]

2. The half-life of ${}^{238}\mathrm{U}$ is $4.468e9$ years. Calculate its decay constant in s^{-1}. [2 marks]

3. A rock sample contains 1.00 g of ${}^{238}\mathrm{U}$ and no lead initially. Assuming a closed system, calculate the mass of ${}^{206}\mathrm{Pb}$ present after $4.468e9$ years (one half-life). [4 marks]

4. The decay series includes the gas ${}_{86}^{222}\mathrm{Rn}$ (radon) with a half-life of 3.82 days. Explain why radon accumulation in basements is a health hazard and how it relates to the uranium content of the soil. [3 marks]

5. In a cloud chamber photograph, a track is observed that is thick, straight, and short (a few cm in air). Another track is thin, tortuous, and much longer. Identify the radiations responsible for each track and explain the differences in appearance based on their interaction with matter. [3 marks]

12 SOLUTIONS TO THE EXERCISES

SOLUTION **Exercise 1 — Multiple choice questions**

1. Answer: (a) α-particles.

   The ionising power of a radiation depends on its charge and its speed (or mass for a given energy): the greater the charge and the slower the particle, the more ion pairs it creates per centimetre of path. The α-particle ($^4_2\mathrm{He}^{2+}$, charge $+2e$, mass $\approx 4\,\mathrm{u}$) is heavy and doubly charged, so it interacts very strongly with matter and produces about 10^4 to 10^5 ion pairs per cm in air. β-particles (charge $-e$, very light) ionise about 100 times less, and γ-rays and X-rays, being uncharged photons, are the least ionising.

2. Answer: (a) $^{A-4}_{Z-2}Y$.

   In α-decay the nucleus emits $^4_2\mathrm{He}$. Conservation of nucleon number gives $A' = A - 4$ and conservation of charge gives $Z' = Z - 2$:

   $$^{A}_{Z}\mathrm{X} \longrightarrow {}^{A-4}_{Z-2}\mathrm{Y} + {}^{4}_{2}\mathrm{He}$$

3. Answer: (c) $\frac{1}{8}$.

   30 days corresponds to $n = \dfrac{30}{10} = 3$ half-lives. The remaining fraction is

   $$\frac{N}{N_0} = \left(\frac{1}{2}\right)^3 = \frac{1}{8}.$$

4. Answer: (c) less than the total mass of the reactants.

   Fission releases energy, and by Einstein's relation $E = \Delta m\, c^2$ this energy corresponds to a mass defect $\Delta m > 0$: the products are lighter than the reactants, the "missing" mass having been converted into kinetic energy of the fragments and radiation.

SOLUTION **Exercise 2 — True or False with justification**

1. False. γ-rays are electromagnetic radiation (photons); they carry no electric charge. The magnetic (Lorentz) force $\vec{F} = q\,\vec{v} \wedge \vec{B}$ is zero for $q = 0$, so γ-rays travel straight through a magnetic field. Only charged radiations (α, β) are deflected.

2. True. The mass defect is defined by

   $$\Delta m = \left[Z\,m_p + (A - Z)\,m_n\right] - m_{\text{nucleus}} > 0.$$

   The nucleus is lighter than the sum of its separated nucleons; the difference Δm corresponds, through $E = \Delta m c^2$, to the binding energy released when the nucleus formed.

3. False. In β^- decay a neutron transforms into a proton inside the nucleus: ${}^{1}_{0}\mathrm{n} \to {}^{1}_{1}\mathrm{p} + {}^{0}_{-1}\mathrm{e} + \bar{\nu}_e$. The mass number A stays constant (correct), but the atomic number increases by 1: ${}^{A}_{Z}\mathrm{X} \longrightarrow {}^{A}_{Z+1}\mathrm{Y} + {}^{0}_{-1}\mathrm{e}$. (It is in β^+ decay that Z decreases by 1.)

4. True. Both nuclei are positively charged and repel each other by the Coulomb force. To bring them within the very short range of the strong nuclear force ($\sim 10^{-15}$ m) so that fusion can occur, they must collide with very high kinetic energy. This requires temperatures of the order of 10^7–10^8 K, as in the core of the Sun or in a tokamak plasma.

SOLUTION **Exercise 3 — Definitions and direct calculations**

1. Definitions.

   (a) Half-life $T_{1/2}$: the time after which half of the radioactive nuclei initially present in a sample have disintegrated (equivalently, the time for the activity to fall to half its initial value).

   (b) Activity A: the average number of disintegrations per unit time in a sample, $A = \lambda N = -\dfrac{dN}{dt}$; SI unit: the becquerel (1 Bq = 1 disintegration per second).

   (c) Binding energy per nucleon B/A: the binding energy of a nucleus divided by its number of nucleons, $\dfrac{B}{A} = \dfrac{\Delta m\, c^2}{A}$. It measures the average energy needed to remove one nucleon; the larger it is, the more stable the nucleus.

2. Half-life from $\lambda = 0.02\ \mathrm{s}^{-1}$:

$$T_{1/2} = \frac{\ln 2}{\lambda} = \frac{0.693}{0.02} \approx 34.7\ \mathrm{s} \approx 35\ \mathrm{s}.$$

3. The activity fell from 800 Bq to 100 Bq, i.e. it was divided by $8 = 2^3$. Hence 12 hours correspond to 3 half-lives:

$$T_{1/2} = \frac{12\ \mathrm{h}}{3} = 4\ \mathrm{h}.$$

4. Balanced equations (conservation of A and Z):

$$^{226}_{88}\mathrm{Ra} \longrightarrow {}^{222}_{86}\mathrm{Rn} + {}^{4}_{2}\mathrm{He} \qquad (\alpha\text{-decay})$$

$$^{14}_{6}\mathrm{C} \longrightarrow {}^{14}_{7}\mathrm{N} + {}^{0}_{-1}\mathrm{e} \qquad (\beta^-\text{-decay})$$

   Check: for radium, $226 = 222 + 4$ and $88 = 86 + 2$; for carbon, $14 = 14 + 0$ and $6 = 7 - 1$. ✓

SOLUTION **Exercise 4 — Radiation identification and properties**

1. A radiation stopped by a single sheet of paper has very low penetrating power: this is characteristic of an **α-particle**. Its deflection is opposite to that of β-particles (which are negative), confirming that it is positively charged: it is the helium nucleus $^{4}_{2}\mathrm{He}^{2+}$, charge $q = +2e = +3.2e-19\ \mathrm{C}$.

2. Completed equations and decay types:
   - $^{238}_{92}\mathrm{U} \longrightarrow {}^{234}_{90}\mathrm{Th} + {}^{4}_{2}\mathrm{He}$: α-decay (A decreases by 4, Z by 2).
   - $^{14}_{6}\mathrm{C} \longrightarrow {}^{14}_{7}\mathrm{N} + {}^{0}_{-1}\mathrm{e}$: β^--decay (A constant, Z increases by 1).
   - $^{214}_{83}\mathrm{Bi} \longrightarrow {}^{214}_{84}\mathrm{Po} + {}^{0}_{-1}\mathrm{e}$: β^--decay (A constant, Z increases by 1).

3. Two properties of each:
   - α-particles: (i) very high ionising power (thick, short, straight tracks in a cloud chamber); (ii) very low penetrating power — stopped by paper or a few cm of air.
   - β^--particles: (i) moderate ionising power, thin and tortuous tracks; (ii) moderate penetrating power — stopped by a few mm of aluminium; they are deflected by magnetic fields (negative charge).

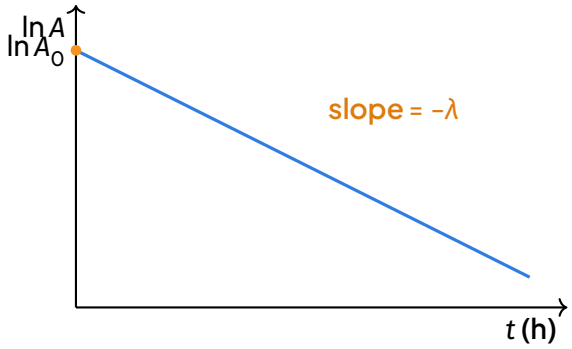
SOLUTION **Exercise 5 — Activity and decay law**

1. Decay constant:
$$\lambda = \frac{\ln 2}{T_{1/2}} = \frac{0.693}{6 \times 3600\ \mathrm{s}} = \frac{0.693}{21600} \approx 3.21e-5\ \mathrm{s}^{-1}.$$

2. From 8:00 am to 2:00 pm, $\Delta t = 6\ \mathrm{h} = 1$ half-life. The activity is halved:
$$A = \frac{A_0}{2} = \frac{1.6e6\ \mathrm{Bq}}{2} = 8.0e5\ \mathrm{Bq}.$$

3. Using the decay law $A = A_0 e^{-\lambda t}$ with $A = 1.0e5\ \mathrm{Bq}$:
$$t = \frac{1}{\lambda} \ln \frac{A_0}{A} = \frac{T_{1/2}}{\ln 2} \ln \frac{1.6 \times 10^6}{1.0 \times 10^5} = \frac{6\ \mathrm{h}}{0.693} \times \ln 16 = \frac{6}{0.693} \times 2.773 \approx 24\ \mathrm{h}.$$
   (Equivalently: $1.6 \times 10^6 / 10^5 = 16 = 2^4$, so 4 half-lives = 24 h.)

4. Since $A = A_0 e^{-\lambda t}$, taking logarithms: $\ln A = \ln A_0 - \lambda t$. The graph of $\ln A$ versus t is a straight line of slope $-\lambda$ and y-intercept $\ln A_0$.



Graph of $\ln A$ against time: a straight line of slope $-\lambda$ and intercept $\ln A_0$.

SOLUTION **Exercise 6 — Mass defect and binding energy**

1. The helium-4 nucleus contains $Z = 2$ protons and $A - Z = 2$ neutrons. Using atomic masses (which automatically account for the electrons) avoids electron-counting errors:

$$\begin{aligned}\Delta m &= \left[2\, m(^1_1\mathrm{H}) + 2\, m_n\right] - m(^4_2\mathrm{He}) \\ &= \left[2(1.007825) + 2(1.008665)\right] - 4.002603 \\ &= 4.032980 - 4.002603 = 0.030377\ \mathrm{u}.\end{aligned}$$

   In kilograms, with $1\ \mathrm{u} = 1.6605 \times 10^{-27}$ kg:

$$\Delta m = 0.030377 \times 1.6605 \times 10^{-27} \approx 5.04e-29\ \mathrm{kg}.$$

2. Total binding energy:

$$B = \Delta m \times 931.5\ \mathrm{MeV} = 0.030377 \times 931.5 \approx 28.30\ \mathrm{MeV}.$$

   In joules ($1\ \mathrm{MeV} = 1.602 \times 10^{-13}$ J):

$$B = 28.30 \times 1.602 \times 10^{-13} \approx 4.53e-12\ \mathrm{J}.$$

   (Check with $E = \Delta m c^2$: $5.04 \times 10^{-29} \times (3.00 \times 10^8)^2 \approx 4.54 \times 10^{-12}$ J. ✓)

3. Binding energy per nucleon:

$$\frac{B}{A} = \frac{28.30\ \mathrm{MeV}}{4} \approx 7.07\ \mathrm{MeV/nucleon}.$$

4. Comment. The value $B/A \approx 7.1$ MeV for ^{4}He is far higher than that of its light neighbours (^{2}H: ≈ 1.1 MeV; ^{3}He: ≈ 2.6 MeV). On the Aston curve, helium-4 sits on a sharp local peak: it is exceptionally stable for such a light nucleus. This explains why α-particles are emitted intact in radioactivity and why fusion of light nuclei into helium releases so much energy.

SOLUTION **Exercise 7 — Nuclear fission energy calculation**

1. Conservation checks.
   - Nucleons: reactants 235 + 1 = 236; products 94 + 140 + 2(1) = 236. ✓
   - Protons: reactants 92 + 0 = 92; products 38 + 54 + 0 = 92. ✓
2. Mass defect:

$$m_{\text{reactants}} = 235.04393 + 1.008665 = 236.052595\ \text{u}$$
$$m_{\text{products}} = 93.91536 + 139.92164 + 2(1.008665) = 235.854330\ \text{u}$$
$$\Delta m = 236.052595 - 235.854330 = 0.198265\ \text{u}.$$

3. Energy released per fission:

$$E = \Delta m \times 931.5 = 0.198265 \times 931.5 \approx 184.7\ \text{MeV} \approx 185\ \text{MeV}.$$

   In joules:

$$E = 184.7 \times 1.602 \times 10^{-13} \approx 2.96e-11\ \text{J}.$$

4. Number of nuclei in 1 g of ^{235}U:

$$N = \frac{m}{M} N_A = \frac{1.00\ \text{g}}{235\ \text{g mol}^{-1}} \times 6.022 \times 10^{23} \approx 2.56 \times 10^{21}\ \text{nuclei}.$$

   Total energy:

$$E_{\text{tot}} = N \times E = 2.56 \times 10^{21} \times 2.96 \times 10^{-11} \approx 7.6e10\ \text{J} \approx 76\ \text{GJ}.$$

5. Comparison with petrol (calorific value $\approx 47\ \text{MJ kg}^{-1} = 4.7e4\ \text{J g}^{-1}$):

$$m_{\text{petrol}} = \frac{E_{\text{tot}}}{4.7 \times 10^4\ \text{J g}^{-1}} = \frac{7.6 \times 10^{10}}{4.7 \times 10^4} \approx 1.6 \times 10^6\ \text{g} \approx 1.6\ \text{t}.$$

   Conclusion: 1 g of ^{235}U releases roughly the same energy as about 1.6 tonnes of petrol — nuclear fuel is about a million times more concentrated in energy than chemical fuel.

SOLUTION **Exercise 8 — Carbon-14 dating**

1. Principle. Cosmic rays striking the upper atmosphere produce neutrons which convert nitrogen-14 into radioactive carbon-14: $^{14}_{7}\text{N} + {}^{1}_{0}\text{n} \longrightarrow {}^{14}_{6}\text{C} + {}^{1}_{1}\text{H}$. The ^{14}C oxidises to CO_2, enters the carbon cycle (photosynthesis → plants → animals), so living organisms maintain a constant $^{14}\text{C}/{}^{12}\text{C}$ ratio and a constant activity per gram of carbon. When the organism dies, it stops exchanging carbon with the atmosphere; the ^{14}C

then decays with $T_{1/2} = 5730$ years. Measuring the remaining activity and applying $A = A_0 e^{-\lambda t}$ gives the time elapsed since death.

2. The activity is 25% = $\frac{1}{4} = \left(\frac{1}{2}\right)^2$ of the living value, so two half-lives have elapsed:

$$t = 2 \times 5730 = 11460 \text{ years} \approx 1.15e4 \text{ years}.$$

3. From $A = A_0 e^{-\lambda t}$, $t = \frac{1}{\lambda} \ln \frac{A_0}{A}$. Differentiating with respect to A:

$$\Delta t = \frac{1}{\lambda} \frac{\Delta A}{A} = \frac{T_{1/2}}{\ln 2} \times 0.02 = \frac{5730}{0.693} \times 0.02 \approx 165 \text{ years}.$$

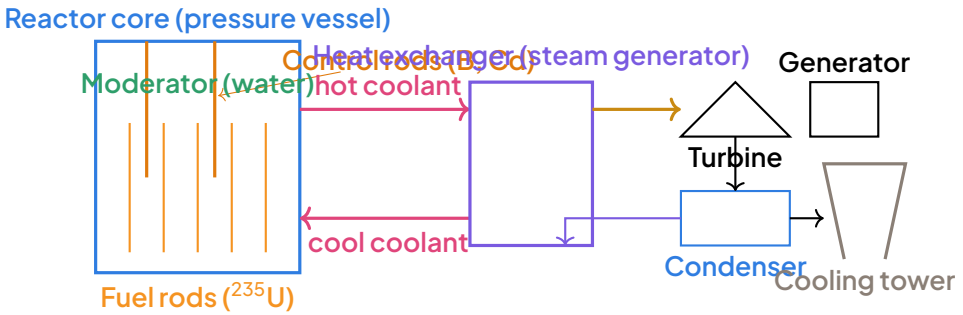
So the age is $t = (11460 \pm 165)$ years (from the activity uncertainty alone).

4. After about 50,000 years, i.e. ~ 9 half-lives, the remaining fraction is $\left(\frac{1}{2}\right)^9 \approx 0.2\%$. The activity is then so weak that it is comparable to background radiation and contamination; the counting uncertainty becomes enormous and the measured "age" is unreliable. Other methods (K–Ar, U–Pb dating) are used for older samples.

SOLUTION **Exercise 9 — Structured essay: Nuclear Reactors and Energy in Cameroon (20 marks)**

Indicative mark scheme and model answer.

1. Labelled diagram [6 marks] — 1 mark per correctly placed and labelled group of components (core + fuel rods; moderator; control rods; coolant circuit; heat exchanger; turbine + generator + condenser + cooling tower).



Schematic of a pressurised water reactor (PWR) and its power conversion circuit.

2. Moderator and control rods [4 marks].

- Moderator (water, heavy water or graphite): slows the fast neutrons released by fission down to thermal energies (≈ 0.025 eV) by elastic collisions. Why slow

them? The probability (cross-section) of ^{235}U capturing a neutron and fissioning is hundreds of times greater for slow (thermal) neutrons than for fast ones; slowing them sustains the chain reaction efficiently. [2 marks]

- Control rods (boron or cadmium, strong neutron absorbers): inserted or withdrawn to adjust the neutron population and hence the reaction rate; fully inserted, they shut the reactor down. [2 marks]

3. Fission and chain reaction [4 marks]. A thermal neutron is absorbed by ^{235}U, forming unstable ^{236}U, which splits into two medium-mass fragments plus 2–3 neutrons and ≈ 200 MeV of energy, e.g.

$$^{235}_{92}\mathrm{U} + {}^{1}_{0}\mathrm{n} \longrightarrow {}^{94}_{38}\mathrm{Sr} + {}^{140}_{54}\mathrm{Xe} + 2\,{}^{1}_{0}\mathrm{n}.$$

The emitted neutrons, once moderated, can trigger further fissions: this is the **chain reaction**. If on average exactly one neutron per fission causes a new fission, the reaction is self-sustaining at constant power (critical state). The **critical mass** is the minimum mass of fissile material needed so that neutron losses through the surface do not quench the reaction. [1 mark per concept: fission process, neutron multiplication, chain reaction, critical mass.]

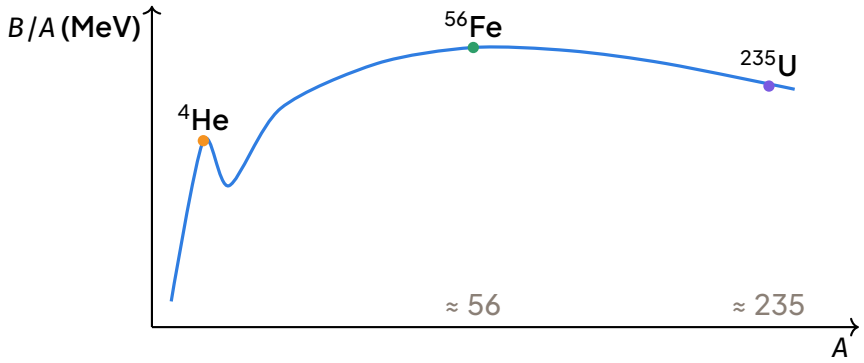
4. Advantages and risks for Cameroon [6 marks] — any two of each, well argued:

- Advantages: (i) enormous energy density — a small quantity of fuel supplies large cities and industry, complementing hydroelectricity (which suffers in dry seasons); (ii) no CO_2 emissions during operation, helping preserve Cameroon's forests and climate commitments; (iii) stable, continuous base-load power unlike seasonal hydro.
- Risks/disadvantages: (i) radioactive waste remains dangerous for thousands of years and Cameroon has no disposal facility; (ii) risk of accidents and need for highly specialised personnel, strict regulation and large capital investment; (iii) reactors need reliable cooling water and siting away from seismic/volcanic zones (e.g. Mount Cameroon region).

Examiner's expectations: a clear, correctly labelled diagram; precise vocabulary (thermal neutrons, cross-section, criticality); balanced discussion linked to the Cameroonian context, not a generic list.

SOLUTION **Exercise 10 — Binding energy curve, fission and fusion (15 marks)**

1. Aston curve [4 marks] — 1 mark for axes, 1 for shape (steep rise, broad maximum near Fe, slow decrease), 1 for the three positions, 1 for the He-4 peak.



Binding energy per nucleon versus nucleon number (Aston curve).

2. Qualitative explanation [4 marks].

   (a) Fission: ^{235}U lies on the descending part of the curve ($B/A \approx 7.6$ MeV). When it splits into two medium-mass nuclei ($A \approx 90$–140), the products lie higher on the curve ($B/A \approx 8.5$ MeV), i.e. they are more tightly bound. The increase in total binding energy is released as kinetic energy and radiation: $E = (B/A)_{\text{products}} - (B/A)_{\text{reactants}} > 0$. [2 marks]

   (b) Fusion: very light nuclei (^{2}H, ^{3}H) lie very low on the curve ($B/A \approx 1$–3 MeV). Fusing them into ^{4}He ($B/A \approx 7.1$ MeV) moves the nucleons up the curve to a much more bound state, releasing the difference as energy. [2 marks]

   In both cases, energy is released because the products are more tightly bound (greater B/A) than the reactants.

3. Energy of the D–T fusion [4 marks].

$$m_{\text{reactants}} = 2.014102 + 3.016049 = 5.030151\ \text{u}$$
$$m_{\text{products}} = 4.002603 + 1.008665 = 5.011268\ \text{u}$$
$$\Delta m = 5.030151 - 5.011268 = 0.018883\ \text{u}$$
$$E = 0.018883 \times 931.5 \approx 17.6\ \text{MeV}.$$

4. Difficulties of controlled fusion [3 marks] — any two:

   - Reaching and maintaining temperatures of $\sim 10^8$ K so that nuclei can overcome Coulomb repulsion — no material container can hold such a plasma (magnetic or inertial confinement is needed).
   - Maintaining a sufficient plasma density and confinement time (Lawson criterion) for more energy to be produced than is injected.
   - The large energy input currently exceeds the energy output: a net energy gain in a continuously operating power reactor is not yet industrially achieved.

SOLUTION **Exercise 11 — Comprehensive radioactivity problem (15 marks)**

1. Number of α and β^- decays [3 marks]. Let n_α and n_β be the numbers of decays. Conservation of nucleon number:

$$238 = 206 + 4n_\alpha \implies n_\alpha = \frac{238 - 206}{4} = 8.$$

Conservation of charge (each α removes 2 units of Z, each β^- adds 1):

$$92 = 82 + 2n_\alpha - n_\beta \implies n_\beta = 82 + 16 - 92 = 6.$$

8 α-particles and 6 β^--particles are emitted per atom.

2. Decay constant [2 marks]. Convert the half-life to seconds:

$$T_{1/2} = 4.468 \times 10^9\ \text{y} \times 365.25 \times 24 \times 3600\ \text{s} \approx 1.410 \times 10^{17}\ \text{s}.$$

$$\lambda = \frac{\ln 2}{T_{1/2}} = \frac{0.693}{1.410 \times 10^{17}} \approx 4.92e-18\ \text{s}^{-1}.$$

3. Mass of lead after one half-life [4 marks]. After one half-life, half of the uranium atoms have decayed: the number of decayed atoms is

$$N_{\text{decayed}} = \frac{1}{2} \times \frac{1.00\ \text{g}}{238\ \text{g mol}^{-1}} \times 6.022 \times 10^{23} \approx 1.265 \times 10^{21}\ \text{atoms}.$$

Each decayed ^{238}U atom ends as one ^{206}Pb atom, so

$$m_{\text{Pb}} = \frac{N_{\text{decayed}}}{N_A} \times 206\ \text{g mol}^{-1} = \frac{1.00}{2} \times \frac{206}{238} \approx 0.433\ \text{g}.$$

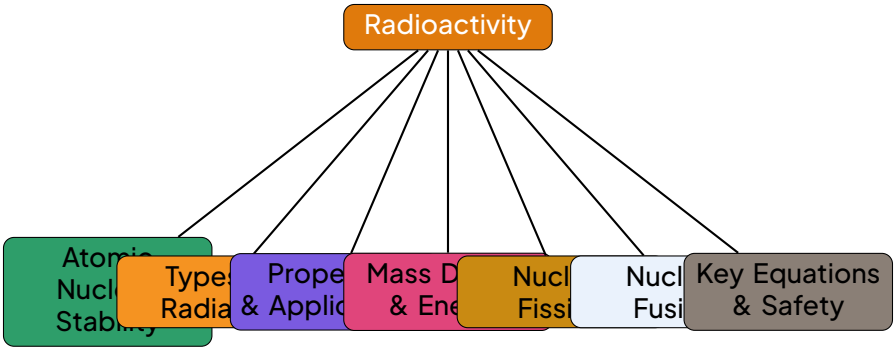
(The remaining uranium mass is 0.50 g; the difference, 0.50 - 0.433 = 0.067 g, is carried away mainly by the 8 emitted α-particles per atom.)

4. Radon hazard [3 marks]. ^{222}Rn is an intermediate member of the ^{238}U series, so soils and rocks containing uranium continuously produce it. Being a chemically inert gas, it diffuses out of the ground and accumulates in poorly ventilated basements. Inhaled radon and especially its solid radioactive daughters deposit in the lungs, where their α-emissions — highly ionising over short ranges — damage lung tissue and increase the risk of lung cancer. The higher the uranium content of the local soil/rock, the greater the radon flux into buildings.

5. Cloud chamber tracks [3 marks].

- Thick, straight, short track: an **α-particle**. Heavy and doubly charged, it ionises intensely (dense track), loses energy rapidly (short range, a few cm in air) and is barely deviated by collisions (straight).
- Thin, tortuous, long track: a **β^--particle** (electron). Very light and singly charged, it ionises weakly (thin track), keeps a long range, and is easily scattered by atomic electrons, giving an irregular, winding path.

13 SUMMARY SHEET



Mind map of the Radioactivity chapter

FICHE BILAN **L'essentiel à connaître pour l'examen**

Key Definitions

- Nucleus: Composed of protons (Z) and neutrons (N), mass number A = Z + N.
- Isotopes: Same Z, different N.
- Radioactivity: Spontaneous disintegration of unstable nuclei with emission of radiation.
- Activity (A): Number of decays per second, unit becquerel (Bq). $A = \lambda N$.
- Decay constant (λ): Probability per unit time of decay.
- Half-life ($T_{1/2}$): Time for activity to halve. $T_{1/2} = \frac{\ln 2}{\lambda}$.
- Mass defect (Δm): Difference between mass of nucleons and actual nuclear mass. $\Delta m = Zm_p + Nm_n - M_{\text{nuc}}$.
- Binding energy (E_b): Energy equivalent of mass defect, $E_b = \Delta mc^2$.
- Nuclear fission: Splitting of heavy nucleus into lighter fragments with energy release.
- Nuclear fusion: Combining light nuclei to form heavier nucleus with energy release.

Laws, Formulas & Theorems

- Radioactive decay law: $N = N_0 e^{-\lambda t}$, $A = A_0 e^{-\lambda t}$.
- Half-life relation: $T_{1/2} = \frac{\ln 2}{\lambda}$.
- Mass-energy equivalence: $E = \Delta mc^2$; 1 u = 931.5 MeV.

- Binding energy per nucleon: $\frac{E_b}{A}$; peak at ^{56}Fe.
- Fission energy: $Q = [M_{\text{initial}} - \sum M_{\text{final}}]c^2$.
- Fusion energy: similar Q-value.
- Radiation types: α (He nucleus), β^- (electron), β^+ (positron), γ (photon).
- Decay equations: ${}^{A}_{Z}X \longrightarrow {}^{A-4}_{Z-2}Y + {}^{4}_{2}He$ (α), ${}^{A}_{Z}X \rightarrow {}^{A}_{Z+1}Y + {}^{0}_{-1}e + \bar{\nu}_e$ (β^-), ${}^{A}_{Z}X \rightarrow {}^{A}_{Z-1}Y + {}^{0}_{+1}e + \nu_e$ (β^+), γ emission no change in A or Z.

Methods

1. Decay problems: Identify N_0, λ or $T_{1/2}$, time t; use $N = N_0 e^{-\lambda t}$.
2. Half-life from data: Plot $\ln A$ vs t; slope = $-\lambda$; $T_{1/2} = \ln 2/\lambda$.
3. Mass defect: Compute $\Delta m = Zm_p + Nm_n - M_{\text{atom}} + Zm_e$ (using atomic masses).
4. Binding energy: $E_b = \Delta m \times 931.5$ MeV.
5. Fission/Fusion Q-value: $Q = (\sum m_{\text{initial}} - \sum m_{\text{final}})c^2$.
6. Radiation identification: Use penetration (paper, Al, Pb) and magnetic deflection.

Common Mistakes

- Confusing activity (Bq) with count rate (cps); count rate = activity × detection efficiency.
- Forgetting to use atomic masses (include electrons) in mass defect calculations.
- Misapplying half-life: $N = N_0(1/2)^{t/T_{1/2}}$ only for integer half-lives; use exponential for exact.
- Mixing up β^- and β^+ decay equations (change in Z).
- Neglecting neutrinos in β decay energy balance.
- Using $c = 3.00 \times 10^8$ m/s but forgetting to convert MeV to joules when needed.
- Assuming all radiation is equally ionising; α is most ionising, γ least.

GCE Tips

- Definitions: Learn precise wording (e.g., "half-life is the time taken for the activity of a sample to fall to half its initial value").
- Graphs: Be able to sketch N vs t, $\ln N$ vs t, and determine half-life from graph.
- Calculations: Show all steps; keep units consistent (kg, MeV, J). Use 1 u = 1.66×10^{-27} kg = 931.5 MeV.
- Nuclear equations: Balance A and Z; include neutrinos/antineutrinos for β decay.

- Applications: Know uses: α (smoke detectors), β (thickness gauges), γ (sterilisation, tracers), fission (power), fusion (future energy).
- Safety: Time, distance, shielding; α stopped by paper, β by few mm Al, γ by thick Pb.
- Cameroon context: Mention potential for nuclear energy (e.g., uranium deposits in Cameroon) and medical isotopes.

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